Submittal Review Response

Project Name:	***Hilo WWTP Rehabilitation and Replacement Project Phase 1***
Submittal No.:	***11403-001.0***
Date:	***10/1/2025***

Client:	County of Hawaiʻi	**Carollo Project No.:**	203975
Contractor:	Nan, Inc.		
Submittal Name:	Tube In Tube Heat Exchanger		
Reviewed By:	Matheus Tolentino Lauar		

SUBMITTAL REVIEW		
Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.		
Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

1. Acknowledge Contractor comment #1 regarding overall dimensions of the equipment.
2. Acknowledge Contractor comment #2 regarding stamped anchorage design submitted being provided separately when available.
3. Contractor comment #3 regarding the three coating system for structural steel, in lieu of galvanized steel: The approach is acceptable.
4. Contractor comment #4 regarding the insulation inside the heat exchanger enclosure: Manufacturer to provide clarification on how the pipe penetrations and other openings would be sealed to prevent the passage of air, as specified in paragraph 11403-2.03.H.3. Considering that Hilo is humid and rainy, there is a concern to mitigate issues related to moisture and potential for mold. Manufacturer to also clarify about servicing the heat exchanger tubes if wrapped in insulation. Are the insulation blankets removable (and reusable)? Insulation of the enclosure, as described in 11403-2.03.H.3, would be installed on the inner side of the panels, and would not obstruct access to the tubes once panels are removed.
5. Paragraph 11403-2.03.L.2 specified water inlet and outlet to be 4-inch diameter, and although the inlet/outlet size was stated correctly in the manufacturer's 'Equipment Specifications' letter, those connections are called out as 6" in shop drawing 1 (first section on the left). Contractor to confirm size of water inlet/outlet, as specified.

6. Sludge pressure gauge is specified with 3/4" NPT connection, per paragraph 11403-2.03.J. Submittal indicates a 1/4" NPT connection, the same as for the water pressure gauge. Revise pressure gauge for sludge service to have 3/4" NPT connection as indicated.

☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii

Contractor: Nan, Inc. **Project No.:** WW-4705R

Project Name: Hilo WWTP Phase 1 **Submittal Number:** 11403-001.0

Submittal Title: TUBE-IN-TUBE HEAT EXCHANGER ☐ For Information Only

TO: County

From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier			
Spec: 11403	**Paragraph:** 1.03 2.01 2.02 2.03 3.05		
Authored By: Walker		**Date Submitted:** 09/02/2025	

Submittal Certification

Check Either (A) or (B):

☑ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.

☐ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.

Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.

General Contractor's Reviewer's Signature: [signature]

Printed Name and Title: Makoa Ng, QC, Project Engineer

In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.

Firm: **Signature:** **Date Returned:**

PM/CM Office Use

Date Received GC to PM/CM:

Date Received PM/CM to Reviewer:

Date Received Reviewer to PM/CM:

Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/2/2025
SPECIFICATION SECTION # 11403
SPECIFICATION TUBE-IN-TUBE HEAT EXCH
PARAGRAPH 1.03 2.01 2.02 2.03 3.05
DRAWING Walker
SUBCONTRACTOR na
SUPPLIER Walker
MANUFACTURER Walker

CERTIFIED BY CQCM or Designee : [signature]

SECTION 11403

TUBE-IN-TUBE HEAT EXCHANGER

PART 1 GENERAL

1.01 SUMMARY

A. Section includes: ASME BPVC VIII, 316 stainless steel digester sludge tube-in-tube heat exchanger(s) with alloy steel water tubes.

1.02 REFERENCES

A. American Society of Mechanical Engineers (ASME):
   1. B16.5 Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24-Inch Standard.
   2. Boiler Pressure Vessel Code VIII – Rules for Construction of Pressure Vessels Div 1.
   3. SA53 Seamless and Welded, Black and Hot-Dipped Galvanized Steel Pipe.

B. ASTM International (ASTM):
   1. A182 - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service.
   1. A193 - Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications.
   2. A194 - Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts and Bolts for High Pressure or High Temperature Service, or Both.
   2. A240 - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications.
   3. A276 - Standard Specification for Stainless Steel Bars and Shapes.
   4. A312 - Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes.
   3. A351 - Standard Specification for Castings, Austenitic, for Pressure-Containing Parts.
   5. A380 - Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems.
   4. A403 - Standard Specification for Wrought Austenitic Stainless Steel Piping Fittings.
   5. A967 - Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts.
   6. F593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.

1.03 SUBMITTALS

A. Submit as specified in Section 01330 - Submittal Procedures and Section 15050 - Common Work Results for General Process Piping.

B. Product data including materials of construction.

C. Shop drawings: Indicating general assembly, installation instructions, dimensions, components, maintenance clearance requirements, and all piping connections.

D. Calculations:
   1. Hot water and sludge flow rates and pressure loss data.
   2. Heat transfer capacity.
   3. Calculations for anchoring equipment to structures.
      a. Indicate anchor type, layout, number, size, and embedment requirements to resist dead, live, seismic, and other applicable loads specified in Section 01850 - Design Criteria.
      b. Calculations shall be prepared and sealed by a Professional Engineer licensed in the state where the Work will be installed.
      c. Design anchors in accordance with the structural design criteria specified in Section 01850 - Design Criteria, and the requirements of this Section.

2. see submittal comments

E. Product data of other equipment provided with the heat exchanger package (e.g., pressure gauges, ASME safety relief valves, etc.).

F. ASME certification.

G. Quality control submittals.

H. Operation and maintenance manuals.

I. Commissioning submittals:
   1. Provide Manufacturer’s Certificate of Source Testing as specified in Section 01756 - Commissioning.
   2. Provide Manufacturer’s Certificate of Installation and Functionality Compliance as specified in Section 01756 - Commissioning.

J. Project closeout documents:
   1. Provide vendor operation and maintenance manual as specified in Section 01782 - Operation and Maintenance Manuals.

1.04 DELIVERY, STORAGE, AND HANDLING

A. As specified in Section 15050 - Common Work Results for General Process Piping.

1.05 PROJECT CONDITIONS

A. Environmental requirements: As specified in Section 01850 - Design Criteria.

1.06 SEQUENCING AND SCHEDULING

A. Coordinate work with restrictions as specified in Section 01140 - Work Restrictions.

1.07 WARRANTY

A. Provide warranty as specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. One of the following or approved equal:
   1. Walker Process, Type WPE Heat X.
   2. OTI, Model HXt4x8.

B. Note that the manufacturer's standard offering may not meet this specification fully, and updates to the product and/or design and construction of the vessel may be necessary to meet these specifications.

C. The Drawings are based on the Walker Process, Model Type WPE Heat X system by Walker Process Corporation.
   1. Contractor shall be responsible for the cost of any changes that may be required to accommodate alternate model and equipment by other manufacturers, including, but not limited to:
      a. Structural.
      b. Mechanical.
      c. Electrical work.
   2. Contractor shall also pay any additional engineering costs necessary for revisions of Drawings and Specifications required to accommodate alternate equipment.

D. Manufacturers shall have at least 5 years of experience in the design, application, construction, and supply of tube in tube heat exchangers for digester sludge in wastewater treatment plants and shall submit a list of operating installations as evidence of meeting the experience requirement.

2.02 DESIGN AND PERFORMANCE CRITERIA

A. Equipment:
   1. Tube-in-tube heat exchanger(s).
   2. Pressure gauges:
      a. Hot water inlet/outlet.
      b. Sludge inlet/outlet.
   3. Temperature probes:
      a. Hot water inlet/outlet.
      b. Sludge inlet/outlet.
   4. ASME safety relief valves, sized for the specific heat exchanger capacity and design criteria.
      a. Hot water side.
      b. Sludge side.
   5. Anchor bolts.
   6. All other appurtenances required for a complete and operable installation.

2.03 EQUIPMENT

A. Heat exchanger:
   1. The exchanger is a counterflow, tube in a tube type heat exchanger consisting of a series of 6-inch schedule 10 316 stainless steel sludge tubes within 8-inch schedule 40 SA53 alloy steel water tubes.
   2. Both water and sludge tubes shall be removable for inspection and replacement.
   3. Non-counter flowing designs will not be allowed.
   4. The sludge heat exchanger shall be designed for the temperatures indicated in the Design Criteria of this Section.

3. see submittal comments

B. Materials:
   1. All structural steel: in accordance with ASTM A36.
      a. Structural steel shall be galvanized.
   2. Stainless steel pipe: in accordance with ASTM A312.
   3. Alloy steel pipe: in accordance with ASME SA53.
   4. Steel members in contact with liquids, either continuously or intermittently, shall have a minimum thickness of 1/4 inches.
   5. All aluminum shall be type 5052, 6061, or 6063 alloy unless noted.

C. The sludge and water tubes shall be joined by independently removable, neoprene gasketed cast iron end castings designed that any leakage occurring will be to the outside of the exchanger, preventing contamination of the heating water by any material circulated through the sludge tubes.

D. The sludge tube end castings shall be removable for inspection and cleaning of the sludge tubes without draining the water tubes.

E. The sludge inlet and outlet adapters shall be furnished with vent valves to aid in bleeding air from the system and plugs to aid in draining the exchanger.

F. The hot water and sludge inlet and outlets shall be oriented to match the piping shown on the drawings.

G. Lifting rings shall be located in each of the upper four comers of the exchanger frame to aid in handling and placement of the exchanger.

4. see submittal comments

H. Insulation panels:
   1. Enclose all straight run sections of sludge and water tubes within a 16-gauge 316 stainless steel enclosure.
   2. Return beds and inlet and outlet nozzles shall not be enclosed or insulated.
   3. Provide minimum 1-inch thick heat resistant urethane foam insulation on the inside of the stainless steel panel enclosure. Seal the pipe penetrations such that air does not pass through the penetrations.

I. Thermometers:
   1. Sludge and water 2-1/2 inch dial type, gas actuated thermometers mounted in the side panel of the exchanger with capillary tubing to sensing bulbs in thermowells in the sludge and water inlet and outlet castings.

2. This type of mounting allows for removal of the sensing bulb without draining the sludge or water tubes.
3. Range: 40 to 240 degrees Fahrenheit.

J. Sludge pressure gauges:
1. 4-1/2 inch dial.
2. 0 to 60 pounds per square inch black aluminum case.
3. Type 316 stainless steel diaphragm seal.
4. Glycerin filled: 3/4 inches NPT process connection.

K. Water pressure gauge:
1. 4-1/2 inch dial.
2. 0 to 100 pounds per square inch black aluminum case.
3. 1/4 inches NPT process connection.

L. Process connections:
1. Sludge inlet and outlet: 6-inch flat-faced 316 stainless steel flanges drilled in accordance with ASME B16.5 for class 150.
2. Water inlet and outlet: 4-inch flat-faced steel flanges drilled in accordance with ASME B16.5 for class 150.

M. Painting/Coating:
1. Non stainless steel structural components, outer surfaces of steel water tubes (including within the enclosure), and framing and non-aluminum materials shall be coated with a multi-layer coating system, including:
   a. Zinc rich epoxy primer.
   b. Fast cure epoxy mastic intermediate coat.
   c. Polyurethane topcoat.
2. Coatings shall be applied in multiple coats per the paint manufacturer's requirements and to the paint manufacturer's published recommended dry film thicknesses for protecting all steel components against an ocean marine (salty air) and hydrogen sulfide laden atmosphere.
3. Contractor shall field touch up or entirely repaint surface finishes damaged during shipment or installation to original condition using original materials and methods.
4. Prepare surfaces to be painted in accordance with paint manufacturer's recommendations.
5. Color to be heat exchanger manufacturer's standard color.

N. Anchorage and fasteners:
1. Anchor bolts:
   a. All anchor bolts shall be a minimum of 1/2 inch diameter and made of type 316 stainless steel.
   b. The equipment manufacturer shall furnish all 316 stainless steel anchor bolts, nuts, and washers required for the equipment.
2. Fasteners:
   a. All structural fasteners shall be a minimum of 1/2 inch diameter and made of type 316 stainless steel.
   b. The equipment supplier shall furnish all fasteners required for the assembly of the equipment.

PART 3 EXECUTION

3.01 GENERAL

A. Heat exchanger with piping and accessories shall be aligned, connected, and installed at the location indicated on the Drawings and in strict conformance with the manufacturer's written instructions, with due allowance for thermal expansions, supports, anchors, and drainage.
   1. All hot surfaces on equipment and piping shall be insulated as specified in Section 15082 - Piping Insulation unless otherwise noted in this Section.

B. Shop assembly: The equipment specified in this Section shall be completely factory assembled as a single unit.

3.02 EXAMINATION

A. Inspect all components for shipping damage, conformance to specifications, and proper torques and tightness of fasteners, as specified in Section 15050 - Common Work Results for General Process Piping.

3.03 INSTALLATION

A. Install equipment as indicated on the Drawings and as specified in Section 15050 - Common Work Results for General Process Piping and the manufacturer's written installation instructions.

3.04 COMMISSIONING

A. As specified in Section 01756 - Commissioning and this Section.

B. Manufacturer services:
   1. Provide certificates:
      a. Manufacturer’s Certificate of Source Testing.
      b. Manufacturer’s Certificate of Installation and Functionality Compliance.
   2. Manufacturer’s Representative onsite requirements: Require manufacturer’s representative to perform the following services as described below and as specified in Section 01756 - Commissioning. The specified durations are the minimum required time on the job site. Additional services and/or longer durations shall be provided as needed at no cost to the Owner to meet the required quality of work. Work to be done in a minimum of 3 trips:
      a. Installation assistance: As required:
         1) Advise/observe the Contractor on the installation of the equipment.
         2) Provide additional assistance as required.
      a. Installation inspection: 1 trip, 1 day minimum.
      b. Functional Testing: 1 trip, 2 days.
   3. Training: As defined in Section 01756 - Commissioning and this Section.
      a. Operations: 3-hour class, 2 sessions.
      b. Mechanical Maintenance: 3-hour class, 1 session.
      c. Combined Electrical and I&C Maintenance: 2-hour class, 1 session.
   2. Final acceptance checkout: 1 workday (trip may be combined with training).

C. Source testing:
  1. Test as specified in Section 15958 - Mechanical Equipment Testing.
  2. Tube-in-Tube Heat Exchanger:
     a. Test witnessing: Not witnessed.
     b. Conduct Level 1 General Equipment Performance Test.

D. Functional testing:
  1. Tube-in-Tube Heat Exchanger:
     a. Conduct Level 2 General Equipment Performance Test.
     b. Test conformance to operating requirements specified in this Section.

3.05 SCHEDULE

A. Heat exchanger: Operating requirements for digester sludge heat exchanger shall be as scheduled.

Tag Numbers	**Heat Exchanger 1: 11-EXC-2313**	**Heat Exchanger 2: 11-EXC-2323**	**Heat Exchanger 3: 11-EXC-2333(1)**
Fluid to be heated:	Digested Sludge	Digested Sludge	Digested Sludge
Heat transfer capacity:	425,000 BTU/Hour	425,000 BTU/Hour	425,000 BTU/Hour
Sludge flow rate:	160 gallons per minute	160 gallons per minute	160 gallons per minute
Allowable sludge pressure loss:	1 foot W.C.	1 foot W.C.	1 foot W.C.
Hot water flow rate:	100 gallons per minute	100 gallons per minute	100 gallons per minute
Hot water inlet temperature:	150 degrees Fahrenheit	150 degrees Fahrenheit	150 degrees Fahrenheit
Hot water discharge temperature:	141.5 degrees Fahrenheit	141.5 degrees Fahrenheit	141.5 degrees Fahrenheit
Allowable hot water pressure loss:	2.1 foot W.C.	2.1 foot W.C.	2.1 foot W.C.
Maximum water pressure:	50 pounds per square inch	50 pounds per square inch	50 pounds per square inch
Minimum heat exchanger tube diameters:			
Sludge:	6-inch	6-inch	6-inch
Water:	8-inch	8-inch	8-inch
Sludge concentration, percent solids:			
Minimum:	0.5	0.5	0.5
Average:	2	2	2
Maximum:	4	4	4

Tag Numbers	Heat Exchanger 1: 11-EXC-2313	Heat Exchanger 2: 11-EXC-2323	Heat Exchanger 3: 11-EXC-2333[(1)]
Entering sludge temperature:	98 degrees Fahrenheit	98 degrees Fahrenheit	98 degrees Fahrenheit
Exiting sludge temperature:	103.3 degrees Fahrenheit	103.3 degrees Fahrenheit	103.3 degrees Fahrenheit
Number of passes:	1	1	1
Tubes per pass:	6	6	6
Maximum overall length (including bends):	9-feet 0-inches	9-feet 0-inches	9-feet 0-inches
Maximum overall width:	2-feet 6-inches	2-feet 6-inches	2-feet 6-inches
Maximum overall height:	4-feet 6-inches	4-feet 6-inches	4-feet 6-inches
Notes: (1) Equipment is included as part of Additive Alternate No. 1 only.			

END OF SECTION

Approval Submittal

NOT TO BE USED FOR CONSTRUCTION PURPOSES

HILO WWTP
HILO, HI
TWO (2) E-425 HEATX
WALKER PROCESS CONTRACT X250480

Division McNish Corporation

Dedicated to the
Water and Wastewater
Industry

840 North Russell Avenue
Aurora, Illinois
Phone: (630) 892-7921
Fax: (630) 892-7951
E-mail: walker.process@walker-process.com
Website: www.walker-process.com

HILO WWTP
HILO, HI
TWO (2) E-425 E HEATX
WALKER PROCESS CONTRACT X250480

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SUBMITTAL COMMENTS

THE FOLLOWING ITEMS WERE NOTED DURING PREPARATION OF OUR SUBMITTAL PACKAGE. SOME COMMENTS MAY BE TO INFORM OR TO FURTHER EXPLAIN FEATURES OF THE EQUIPMENT SUBMITTED FOR APPROVAL, WHILE OTHER COMMENTS ARE MADE TO REQUEST ACKNOWLEDGEMENT OF COMPLIANCE WITH THE SPECIFICATIONS, OR ACCEPTANCE OF ALTERNATIVES.

NOTE THAT SOME OF THE COMMENTS MAY REQUIRE DIMENSIONS AND ELEVATIONS BE CONFIRMED AND/OR ESTABLISHED. THIS INFORMATION MUST BE ESTABLISHED PRIOR TO OUR COMPLETING THE ENGINEERING DETAILS OF THE EQUIPMENT.

PLEASE ACKNOWLEDGE ALL COMMENTS. NO ACKNOWLEDGEMENT INDICATES ACCEPTANCE OR APPROVAL.

CONTRACTOR – Note comments requiring your attention.

1. Please refer to Drawing 1 for the overall dimensions of the HeatX. Adequate entryways are required to accommodate the HeatX. WPE assumes no costs related to building modifications, if needed.
2. Paragraph 1.03 D 3 requires WPE to submit stamped anchorage design calculations. These will be provided under a separate cover when they are available.
3. Paragraph 2.03 B 1 a requires the structural steel frame of the heat exchanger to be galvanized. We have submitted the exchanger frame to be a 3 shop coat painted system same as the water tubes. We recommend a painted frame because the units will be located in a warm, moist environment very close to the ocean and there is a high likelihood that galvanic corrosion will occur in the areas of contact between the stainless-steel outer panels and the galvanized frame. A painted frame will have a longer life. Please confirm this is acceptable.
4. Paragraph 2.03 H requires a minimum 1” thick urethane foam insulation inside the heat exchanger enclosure. WPE Heat Exchanger tubes come wrapped in a 2” Knauf KN-75 fiberglass insulating blanket which is superior to the specified insulation.

HILO WWTP
HILO, HI
TWO (2) E-425 HEATX
WALKER PROCESS EQUIPMENT CONTRACT X250480

EQUIPMENT SPECIFICATIONS

Specification Section: 11403 TUBE-IN-TUBE HEAT EXCHANGER

Application: Sludge Heating

Number of Units: 2

Exchanger Heat Transfer Rating: 425,000 BTU / hr

Water Circulation: 100 GPM at 150°F with 1.7 ft. W.C. Head Loss per pass
Sludge Circulation: 160 GPM at 98°F with 0.3 ft. W.C. Head Loss per pass

Exchanger Maximum Operating Pressure: 50 psi

Exchanger:

The exchanger is a counterflow, tube in a tube type exchanger consisting of a series of 6” ASME SA-312 Schedule 10 Type 316 SS Pipe sludge tubes within 8” ASME SA-53B Schedule 40 Black Steel Pipe water tubes. Both water and sludge tubes are removable for inspection and replacement.

The sludge and water tubes are joined by independently removable, nitrile gasketed cast iron end castings so designed that any leakage occurring will be to the outside of the exchanger thus preventing contamination of the heating water by any material circulated through the sludge tubes. The sludge tube end castings can be removed for inspection and cleaning of the sludge tubes without draining the water tubes.

The sludge and water return bends are furnished with vent valves to aid in bleeding air from the system and plugs to aid in draining the exchanger. The sludge and water inlet/outlet adapters are furnished with plugs to aid with draining the exchanger.

All sludge passageways are sized to pass 4 1/2" diameter spheres.

The tubes are supported by 3/4” thick steel end plates with structural steel members serving as supports and framing for the top and side 16-gauge exterior stainless-steel paneling. The entire sludge-water tube bundle is wrapped with 2” thick fiberglass blanket insulation.

The total heated surface area based on the sludge tube exterior diameter is 70.9 ft^2.

The heat exchanger is constructed to ASME Rules for Construction for unfired pressure vessels. The exchanger is inspected by an ASME qualified inspector and the ASME “U” designator is stamped on the heat exchanger nameplate.

Pressure Relief Valves:

The sludge and water passes are each protected by an F-82 ¾" CASH ACME, ASME rated pressure relief valve set for 50 PSI.

Process Connections:

The sludge inlet and outlet are 6" flat faced cast iron flanges drilled in accordance with ANSI B16.1 for 125 lb. flanges.

The water inlet and outlet are 4" flat faced cast iron flanges drilled in accordance with ANSI B16.1 for 125 lb. flanges.

Thermometers:

2 1/2" dial type, gas actuated thermometers mounted in the side panel of the exchanger with capillary tubing to sensing bulbs in thermowells in the sludge and water inlet and outlet castings. This type of mounting allows for removal of the sensing bulb without draining the sludge or water tubes.

Water:	40° to 240°F
Sludge:	0° to 150°F

Pressure Gauges:

4 ½" dial, 60 psi black phenolic case, glycerin filled, 1/4" NPT process connection. Each pressure gauge with diaphragm is isolated with 316 stainless steel ball valves.

Water:	Bronze/Brass Diaphragm seal, 0 to 100 psi
Sludge:	316SS Diaphragm Seal 0 to 60 psi

Fasteners:

316 Stainless Steel

Anchorage:

316 Stainless Steel, cast-in-place

Painting:

See separate Shop Preparation and Painting Specifications.

Spare Parts: None

Special Tools:

None required and none furnished by Walker Process.

SHOP PREPARATION AND PAINTING SPECIFICATIONS

The outer panels, sludge tubes and sludge return bends will be 316 stainless steel and will not require surface preparation.

All other fabricated ferrous parts are to be cleaned to SSPC-SP 7 and coated with:
Manufacturer: Sherwin-Williams

			DFT Per Coat (mils)	
Coat	**Grade**	**Color**	**Min**	**Max**
Prime	Zinc Clad II	Blue	2.0	4.0
Inter	Macropoxy 646 Epoxy	Red Oxide	5.0	7.0
Finish	Acrolon 218 HS Acrylic Polyurethane	ANSI #70 Gray	2.0	4.0

NOTES:

Coatings and/or surface preparations shown above are in full compliance with the contract documents, or our interpretation of them. Lacking said contract documents on coatings and/or surface preparation specifications, we have elected to use Walker Process Equipment standard primer and/or surface preparation specifications. In either case, the contractor is responsible for the compatibility of primer coat and field applied finish coatings.

Purchased components: Walker Process Equipment does not repaint purchased components supplied to walker process equipment with a finished paint system from the manufacturer unless specifically noted above to the contrary.

All field touch up of mars, scratches, bruises, etc., received by equipment during shipment, handling, storage or erection are to be by other than Walker Process Equipment .

All finish coats are not by Walker Process Equipment unless otherwise stated above. Finish coats must be the same type and by the same paint manufacturer as the prime coat to ensure optimum compatibility and avoiding paint warranty invalidation.

Application of field coating(s) shall be in strict compliance with the coating manufacturer's recommendations. Prior to application of field coat(s), the field painter must ensure that the maximum recoat time for the shop coating, as set forth by the coating manufacturer, will not be exceeded. If the maximum recoat time will be exceeded, the field painter shall consult the manufacturer of the shop coating for necessary preparation prior to applying subsequent top coats. Application of field coatings shall be construed as the field painter's acceptance of both the shop surface preparation and the shop applied coating(s). Walker Process Equipment will not accept any backcharges related to either the shop surface preparation or the shop applied coating(s) after field coatings are applied.
Evaluation of dry coating thickness complies with the requirements of industry standard SSPC-PA2, "Paint Application Specification No. 2, Measurement of Dry Coating Thickness With Magnetic Gages".

Walker Process Equipment applies no shop coatings to aluminum, stainless steel or other nonferrous metals, or to galvanized metal unless specifically designated.

Prior to field sandblasting or painting any of the structures, the painting contractor shall take proper steps to protect all gears, motors, drives, mixers, shafting, electrical controls and any other miscellaneous items of equipment that could be damaged by sandblasting or painting. Where open gears such as spur gears on clarifiers are installed, suitable protection must be provided on all areas to prevent any entrance of the sand or paint into the component areas or on to any seals

WARNING: Painting can damage seals and plug breather vents on the drive units. Walker Process Equipment will not be responsible for leaks or loss of lubricant due to field applied paint on seals and or vents.

Unless noted to the contrary, all pipes, tubes, etc., fourteen (14) inches in diameter and larger are to have both interior and exterior surfaces prepared and coated in accordance with these specifications. For pipes, tubes, etc., smaller than fourteen (14) inches in diameter, surface preparation and coating is only to extend into the ends of the pipes, tubes, etc., as far as the gun will reach without inserting the gun within the pipe or tube.

Protective & Marine Coatings
PRODUCT DATA SHEET

ZINC CLAD® II (85)
INORGANIC ZINC RICH COATING

Revised: March 19, 2019

PRODUCT DESCRIPTION

ZINC CLAD II (85) is a solvent-based two-package, inorganic ethyl silicate, zinc-rich coating.
- 85% zinc content in dry film
- Coating self-heals to resume protection if damaged
- Provides cathodic/sacrificial

INTENDED USES

- For use over properly prepared blasted steel
- As a one-coat maintenance coating or as a permanent primer for severely corrosive environments (pH range 5-9)
- Economical replacement for galvanizing with similar performance
- Ideal for application at low temperatures or service at high temperatures and/or humidity conditions
- Where abrasion resistance and hardness is required
- Not recommended for severe acid or alkali exposure

PRODUCT DATA

Finish: Flat

Colors: Gray-Green

Volume Solids: 62% ± 2%, ASTM D2697, mixed

VOC (mixed): <500 g/L; 4.17 lb/gal

Mix Ratio: 2 components, premeasured
5 gallons (18.9L) mix

Typical Thickness:

Recommended Spreading Rate per coat:

	Minimum	Maximum
Wet mils (microns)	**3.5** (88)	**6.5** (163)
Dry mils (microns)	**2.0** (50)	**4.0** (100)
~Coverage sq ft/gal (m^2/L)	**248** (6.1)	**492** (12.2)
Theoretical coverage **sq ft/gal** (m^2/L) @ 1 mil / 25 microns dft	**995** (24.3)	

Dry film thickness in excess of 6.0 mils (150 microns) per coat is not recommended.

Shelf Life: Part E: 9 months, unopened
Part F: 24 months, unopened
Store indoors at 40°F (4.5°C) to 100°F (38°C).

Flash Point: 55°F (13°C), PMCC, mixed

Reducer/Clean Up:
Below 80°F (27°C): Xylene
Above 80°F (27°C): Reducer #58 or High Flash Naphtha - 100

Weight: 20.9 ± 0.2 lb/gal ; 2,5 Kg/L, mixed

Average Drying Times @ 5.0 mils wet (125 microns):

	55°F (13°C)	77°F (25°C) *50% RH*	100°F (38°C)
Touch:	30 minutes	20 minutes	15 minutes
Handle:	3 hours	1-2 hours	20 minutes
Recoat:			
minimum:	48 hours	18 hours	18 hours
maximum:	n/a	n/a	n/a
Cure to service:			
atmospheric:	7 days	7 days	7 days
immersion:	14 days	14 days	14 days
Pot Life:	18 hours	8 hours	6 hours

Note: high humidity will shorten pot life

SURFACE PREPARATION

Surface must be clean, dry, and in sound condition. Remove all oil, dust, grease, dirt, loose rust, and other foreign material to ensure adequate adhesion.

Zinc rich coatings require direct contact between the zinc pigment in the coating and the metal substrate for optimum performance.

Minimum recommended surface preparation:
Iron & Steel: Atmospheric: SSPC-SP6 / NACE 3 / ISO8501-1:2007 Sa 2, 2 mil (50 micron) profile
Immersion: SSPC-SP10 / NACE 2 / ISO8501-1:2007 Sa 2.5, 2 mil (50 micron) profile

Protective & Marine Coatings

PRODUCT DATA SHEET

ZINC CLAD® II (85)

INORGANIC ZINC RICH COATING

APPLICATION

Airless Spray
(use Teflon packings and continuous agitation)

Pressure........................1800-2000 psi (124-137 bar)
Hose..............................3/8" ID (9.5 mm)
Tip..................................017"-.021" (0.43-0.53 mm)
Reduction......................As needed up to 10% by volume

Conventional Spray
(continuous agitation required)

Gun ....Binks 95
Fluid Nozzle 66
Air Nozzle......................63PB
Atomization Pressure.....30-40 psi (2-2.7 bar)
Fluid Pressure................10-20 psi (0.7-1.4 bar)
Reduction......................As needed up to 10% by volume

Keep pressure pot at level of applicator to avoid blocking of fluid line due to weight of material. Blow back coating in fluid line at intermittent shutdowns, but continue agitation at pressure pot.

Brush

Brush.............................For touch-up only

If specific application equipment is not listed above, equivalent equipment may be substituted.

RECOMMENDED SYSTEMS

Dry Film Thickness / ct.		Mils	(Microns)
Steel, Zinc Primer/Finish, Immersion or Atmospheric			
1 Ct.	Zinc Clad II (85)	2.0-4.0	(50-100)
Steel, Epoxy Topcoat, Atmospheric			
1 Ct.	Zinc Clad II (85)	2.0-4.0	(50-100)
1 Ct.	Macropoxy 646	5.0-10.0	(125-250)
Steel, Epoxy/Urethane Topcoat, Atmospheric			
1 Ct.	Zinc Clad II (85)	2.0-4.0	(50-100)
1 Ct.	Macropoxy 646	5.0-10.0	(125-250)
1 Ct.	Acrolon 7300	2.0-4.0	(50-100)
Steel, Epoxy/Urethane Topcoat, Atmospheric			
1 Ct.	Zinc Clad II (85)	2.0-4.0	(50-100)
1 Ct.	Macropoxy 267	5.0	(250)
1 Ct.	Acrolon 7300	2.0-4.0	(50-100)
Steel, Epoxy/Polysiloxane Topcoat, Atmospheric			
1 Ct.	Zinc Clad II (85)	2.0-4.0	(50-100)
1 Ct.	Macropoxy 267	5.0	(250)
1-2 Cts.	Sher-Loxane 800	2.0-4.0	(50-100)

The systems listed above are representative of the product's use, other systems may be appropriate.

WARRANTY

APPLICATION CONDITIONS

Temperature:

Air and surface:	0°F (-17°C) minimum, 120°F (49°C) maximum
Material:	40°F (4.5°C) minimum At least 5°F (2.8°C) above dew point
Relative humidity:	40%-90% maximum

Water misting may be required at humidity below 50%

APPROVALS

- Meets AASHTO M-300 specification
- Meets Class B requirements for Slip Coefficient and Creep Resistance, .56
- This product meets specific design requirements for non-safety related nuclear plant applications in Level II, III and Balance of Plant, and DOE nuclear facilities*

* Nuclear qualifications are NRC license specific to the facility

ADDITIONAL NOTES

Mixing Instructions: Thoroughly agitate Binder Part E using low speed continuous air driven agitation. Slowly mix all of Zinc Dust Part F into all of Binder Part E until mixture is completely uniform. After mixing, pour mixture through 30-60 mesh screen. Mixed material must be used within 8 hours. Do not mix previously mixed material with new. If reducer solvent is used, add only after both components have been thoroughly mixed. Continuous agitation of mixture during application is required, otherwise zinc dust will quickly settle out.

Do not tint.

Excessive film build, poor ventilation, and cool temperatures may cause solvent entrapment and premature coating failure.

Any salting on the zinc surface due to weathering exposure must be removed prior to topcoating.

An intermediate coat is recommended to provide uniform appearance of the topcoat.

Oil base, alkyd, epoxy ester, and silicone alkyd topcoats are not recommended.

Topcoats may be applied once 50 MEK double rubs are achieved. No zinc or only slight traces should be visible. Coin hardness test can also be used.

HEALTH AND SAFETY

Refer to the SDS sheet before use.

Published technical data and instructions are subject to change without notice. Contact your Sherwin-Williams representative for additional technical data and instructions.

DISCLAIMER

Protective & Marine Coatings
PRODUCT DATA SHEET

MACROPOXY® 646
FAST CURE EPOXY MASTIC

Revised: October 19, 2021

PRODUCT DESCRIPTION

MACROPOXY 646 Fast Cure Epoxy Mastic is a high solids, high build, fast drying, polyamide epoxy designed to protect steel and concrete in industrial exposures. Ideal for maintenance painting and fabrication shop applications. The high solids content ensures adequate protection of sharp edges, corners, and welds. This product can be applied directly to marginally prepared steel surfaces.

INTENDED USES

- Recommended for marine applications, refineries, offshore platforms, fabrication shops, chemical plants, tank exteriors, power plants, water treatment plants, and mining and minerals industry
- Limited colors are acceptable for immersion use for salt water and fresh water, not acceptable for potable water

PRODUCT DATA

Finish: Semi-Gloss

Colors: Mill White, Black and a wide range of colors available through tinting

Volume Solids: 72% ± 2%, mixed, Mill White

VOC (mixed): <250 g/L; 2.08 lb/gal

Mix Ratio: 1:1 by volume

Typical Thickness:

Recommended Spreading Rate per coat:

	Minimum	Maximum
Wet mils (microns)	**7.0** (175)	**13.5** (338)
Dry mils (microns)	**5.0*** (125)	**10.0** (250)
~Coverage sq ft/gal (m²/L)	**115** (2.9)	**230** (5.8)
Theoretical coverage **sq ft/gal** (m²/L) @ 1 mil / 25 microns dft	**1152** (28.2)	

*May be applied at 3.0-10.0 mils (75-250 microns) dft as an intermediate in a multicoat system.

NOTE: Brush or roll application may require multiple coats to achieve maximum film thickness and uniformity of appearance.

Shelf Life: 36 months, unopened
Store indoors at 40°F (4.5°C) to 110°F (43°C).

Flash Point: 91°F (33°C), TCC, mixed

Reducer/Clean Up: Reducer #111 or Oxsol 100

Weight: 12.9 ± 0.2 lb/gal ; 1.55 Kg/L, mixed, may vary by color

Average Drying Times @ 7.0 mils (175 microns) wet:

	35°F (1.7°C) *50% RH*	77°F (25°C) *50% RH*	100°F (38°C) *50% RH*
Touch:	4-5 hours	2 hours	1.5 hours
Handle:	48 hours	8 hours	4.5 hours
Recoat:			
minimum:	48 hours	8 hours	4.5 hours
maximum:	1 year	1 year	1 year
Cure to service:			
atmospheric:	10 days	7 days	4 days
immersion:	14 days	7 days	4 days

Average Drying Times as intermediate @ 5.0 mils (125 microns) wet:

	35°F (1.7°C)	77°F (25°C)	100°F (38°C)
Touch:	3 hours	1 hour	1 hour
Handle:	48 hours	4 hours	2 hours
Recoat:			
minimum:	16 hours	4 hours	2 hours
maximum:	1 year	1 year	1 year

If maximum recoat time is exceeded, abrade surface before recoating.
Drying time is temperature, humidity, and film thickness dependent.
Paint temperature must be 40°F (4.5°C) minimum.

Pot Life:	10 hours	4 hours	2 hours
Sweat-in-time:	30 minutes	30 minutes	15 minutes

SURFACE PREPARATION

Surface must be clean, dry, and in sound condition. Remove all oil, dust, grease, dirt, loose rust, and other foreign material to ensure adequate adhesion.

Minimum recommended surface preparation:

Iron & Steel:	Atmospheric: SSPC-SP2/3/ ISO8501-1:2007 St 2 or SSPC-SP WJ-3 / NACE WJ-3L Immersion: SSPC-SP10 / NACE 2/ ISO8501-1:2007 Sa 2.5, 2-3 mil (50-75 micron) profile or SSPC-SP WJ-2/NACE WJ-2L
Stainless Steel:	Atmospheric: SSPC-SP16, 1 mil (25 micron) profile
Aluminum & Galvanizing:	SSPC-SP1. If surface has not be weathered for more than 6 months, follow SSPC-SP1 then SSPC-SP16. For fire proofing projects, consult a Sherwin-Williams representative for surface preparation requirements.
Concrete & Masonry:	Atmospheric: SSPC-SP13/NACE 6, or ICRI No. 310.2R CSP 1-3 Immersion: SSPC-SP13/NACE 6-4.3.1
Ductile Iron Pipe:	Atmospheric: NAPF 500-03-03 Power Tool Cleaning Buried & Immersion: NAPF 500-03-04 Abrasive Blast Cleaning Cast Ductile Iron Fittings: NAPF 500-03-05 Abrasive Blast Cleaning

Protective & Marine Coatings

PRODUCT DATA SHEET

MACROPOXY® 646

FAST CURE EPOXY MASTIC

APPLICATION

Airless Spray*

Pump	30:1
Pressure	2800-3000 psi (193-206 bar)
Hose	1/4" ID (6.3 mm)
Tip	.017"-.023" (0.43-0.58 mm)
Filter	60 mesh
Reduction	As needed up to 10% by volume

Conventional Spray*

Gun	DeVilbiss MBC-510
Fluid Tip	E
Air Nozzle	704
Atomization Pressure	60-65 psi (4.1-4.5 bar)
Fluid Pressure	10-20 psi (0.7-1.4 bar)

Brush*

Brush	Nylon/Polyester or Natural Bristle

Roller*

Cover	3/8" woven with solvent resistant core

Plural Component Spray .. Acceptable

*Reduction .... As needed up to 10% by volume

If specific application equipment is not listed above, equivalent equipment may be substituted.

RECOMMENDED SYSTEMS

Dry Film Thickness / ct.		Mils	(Microns)
Steel & Ductile Iron, Immersion & Atmospheric			
2 Cts.	Macropoxy 646	5.0-10.0	(125-250)
Steel, Organic Zinc Primer, Atmospheric			
1 Ct.	Zinc Clad IV (85)	3.0-5.0	(75-125)
1 Ct.	Macropoxy 646	5.0-10.0	(125-250)
Steel, Inorganic Zinc Primer, Atmospheric			
1 Ct.	Zinc Clad II (85)	2.0-4.0	(50-100)
1 Ct.	Macropoxy 646	5.0-10.0	(125-250)
Steel, Organic Zinc/Epoxy/Urethane Topcoat			
1 Ct.	Zinc Clad IV (85)	3.0-5.0	(75-125)
1 Ct.	Macropoxy 646	3.0-10.0	(75-250)
1 Ct.	Acrolon 7300	2.0-4.0	(50-100)
Steel, Inorganic Zinc/Epoxy/Urethane Topcoat			
1 Ct.	Zinc Clad II (85)	2.0-4.0	(50-100)
1 Ct.	Macropoxy 646	3.0-10.0	(75-250)
1 Ct.	Acrolon 7300	2.0-4.0	(50-100)
Steel, Organic Zinc/Epoxy/Polysiloxane Topcoat, Atmospheric			
1 Ct.	Zinc Clad IV (85)	3.0-5.0	(75-125)
1 Ct.	Macropoxy 646	3.0-10.0	(75-250)
1-2 Cts.	Sher-Loxane 800	2.0-4.0	(50-100)
Concrete/Masonry, Smooth, Immersion & Atmospheric			
2 Cts.	Macropoxy 646	5.0-10.0	(125-250)

The systems listed above are representative of the product's use, other systems may be appropriate.

WARRANTY

The Sherwin-Williams Company warrants our products to be free of manufacturing defects in accord with applicable Sherwin-Williams quality control procedures. Liability for products proven defective, if any, is limited to replacement of the defective product or the refund of the purchase price paid for the defective product as determined by Sherwin-Williams. NO OTHER WARRANTY OR GUARANTEE OF ANY KIND IS MADE BY SHERWIN-WILLIAMS, EXPRESSED OR IMPLIED, STATUTORY, BY OPERATION OF LAW OR OTHERWISE, INCLUDING MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

APPLICATION CONDITIONS

Temperature:

Air: 35°F (1.7°C) minimum, 120°F (49°C) maximum
Surface*: 35°F (1.7°C) minimum, 250°F (120°C) maximum
Material: 40°F (4.5°C) minimum
At least 5°F (2.8°C) above dew point

Relative humidity: 85% maximum

*Application to surfaces above 120°F (49°C) is not recommended in VOC Restricted Areas (≤250 g/L). When spraying a surface above 120°F (49°C) in other areas (>250 g/L), please consult with your Sherwin-Williams representative.

APPROVALS

- Suitable for use in USDA inspected facilities
- Acceptable for use in Canadian Food Processing facilities, categories: D1, D2, D3 (Confirm acceptance of specific part numbers/rexes with your SW Sales Representative)
- Conforms to AWWA D102 OCS #5
- Conforms to MPI # 108
- This product meets specific design requirements for non-safety related nuclear plant applications in Level II, III and Balance of Plant, and DOE nuclear facilities*
- Meets Class A requirements for Slip Coefficient, 0.36 @ 6 mils / 150 microns dft (Mill White only)
- Approved intermediate for NEPCOAT System B
- Approved to Norsok M501 system 7B (limited colors)

* Nuclear qualifications are NRC license specific to the facility

ADDITIONAL NOTES

Tint Part A with Maxitoners at 150% strength. Five minutes minimum mixing on a mechanical shaker is required for complete mixing of color.

Tinting is not recommended for immersion service.

Quick-Kick Epoxy Accelerator is acceptable for use. See data page for details.

Acceptable for concrete floors.

Application to surfaces above 120°F (49°C) is not recommended in VOC Restricted Areas (≤250 g/L). When spraying a surface above 120°F (49°C) in other areas (>250 g/L), please consult with your Sherwin-Williams representative. Spray apply only. Product will produce an orange peel appearance when applied at elevated temperatures.

Topcoating: It is recommended to apply a thinned-down, low wet film thickness mist coat over zinc rich primers to help avoid outgassing. Allow it to tack up and seal the surface. Then apply a full wet film thickness coat as directed.

Mix contents of each component thoroughly with low speed power agitation. Make certain no pigment remains on the bottom of the can. Then combine one part by volume of Part A with one part by volume of Part B. Thoroughly agitate the mixture with power agitation. Allow the material to sweat-in as indicated prior to application. Re-stir before using.

HEALTH AND SAFETY

Refer to the SDS sheet before use.
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DISCLAIMER

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Protective & Marine Coatings

ACROLON™ 218 HS
ACRYLIC POLYURETHANE

PART A	B65-600	GLOSS SERIES
PART A	B65-650	SEMI-GLOSS SERIES
PART B	B65V600	HARDENER

Revised: July 22, 2021

PRODUCT INFORMATION

5.22

PRODUCT DESCRIPTION

ACROLON 218 HS is a polyester modified, aliphatic, acrylic polyurethane formulated specifically for in-shop applications. Also suitable for industrial applications. A fast drying, urethane that provides color and gloss retention for exterior exposure.

- Can be used directly over organic zinc rich primers (epoxy zinc primer and moisture cure urethane zinc primer)
- Color and gloss retention for exterior exposure
- Fast dry
- Outstanding application properties

PRODUCT CHARACTERISTICS

Finish: Gloss or Semi-Gloss

Color: Wide range of colors available

Volume Solids: 65% ± 2%, mixed, may vary by color

Weight Solids: 78% ± 2%, mixed, may vary by color

VOC (EPA Method 24):

	Unreduced:	<300 g/L; 2.5 lb/gal
mixed	Reduced 10% with R7K15:	<340 g/L; 2.8 lb/gal
mixed	Reduced 9% with MEK, R6K10:	<340 g/L; 2.8 lb/gal

Mix Ratio: 6:1 by volume, 1 gallon or 5 gallon mixes premeasured components

Recommended Spreading Rate per coat:

	Minimum		Maximum	
Wet mils (microns)	**4.5**	(112.5)	**9.0**	(225)
Dry mils (microns)	**3.0**	(75)	**6.0**	(150)
~Coverage sq ft/gal (m^2/L)	**175**	(4.3)	**346**	(8.5)
Theoretical coverage **sq ft/gal** (m^2/L) @ 1 mil / 25 microns dft	**1040**	(25.5)		

NOTE: Brush or roll application may require multiple coats to achieve maximum film thickness and uniformity of appearance.

Drying Schedule @ 6.0 mils wet (150 microns):

	@ 35°F/1.7°C	@ 77°F/25°C *50% RH*	@ 120°F/49°C
To touch:	4 hours	1 hour	20 minutes
To handle:	18 hours	9 hours	4 hours
To recoat:			
minimum:	18 hours	8 hours	6 hours
maximum:	3 months	3 months	3 months
To cure:	14 days	7 days	5 days
Pot Life: (reduced 5% with Reducer R7K15)	4 hours	2 hours	45 minutes
Sweat-in-Time:		None	

Drying time is temperature, humidity, and film thickness dependent.
Paint temperature must be at least 40°F (4.5°C) minimum.

Shelf Life: Part A* - 36 months, unopened
Part B - 24 months, unopened
Store indoors at 40°F (4.5°C) to 100°F (38°C).

*Aluminum (Part A, Rex # B65SW655) has a shelf life of 24 months.

Flash Point: 55°F (13°C), Seta, mixed

Reducer/Clean Up:

Spray: Reducer R7K15, MEK R6K10, R7K111, Reducer #58

Brush / Roll: Reducer #132, Reducer #58, R7K111

RECOMMENDED USES

Specifically formulated for in-shop applications.
For use over prepared metal and masonry surfaces in industrial environments such as:

- Structural steel
- Rail cars and locomotives
- Conveyors
- Bridges
- Tank exteriors
- Pipelines
- Ships
- Wind Towers - onshore and offshore
- Offshore platforms - exploration and production
- Suitable for use in USDA inspected facilities
- Conforms to AWWA D102 Outside Coating Systems #4 (OCS-4), #5 (OCS-5) & #6 (OCS-6)
- Conforms to MPI# 72 and MPI# 174
- Acceptable for use in high performance architectural applications
- Acceptable for use over and/or under Loxon S1 and Loxon H1 Caulking
- A component of INFINITANK
- Over FIRETEX® hydrocarbon systems
- Suitable for use in the Mining & Minerals Industry
- Approved topcoat for NEPCOAT System B

PERFORMANCE CHARACTERISTICS

Substrate*: Steel

Surface Preparation*: SSPC-SP10/NACE 2

System Tested*:

1 ct. Macropoxy 646 @ 6.0 mils (150 microns) dft
1 ct. Acrolon 218 HS Gloss @ 4.0 mils (100 microns) dft

*unless otherwise noted below

Test Name	Test Method	Results
Abrasion Resistance[1]	ASTM D4060, CS17 wheel, 1000 cycles, 1 kg load	43 mg loss
Adhesion[3]	ASTM D4541	1976 psi
Corrosion Weathering[3]	ASTM D5894, 27 cycles, 9072 hours	Rating 10 per ASTM D610, for rusting; Rating 10 per ASTM D714, for blistering
Direct Impact Resistance[1]	ASTM D2794	70 in. lb.
Dry Heat Resistance[1]	ASTM D2485, Method A	200°F (93°C)
Flexibility[1]	ASTM D522, 180° bend, 1/8" mandrel	Passes
Humidity Resistance[2]	ASTM D4585, 100°F (38°C), 1500 hours	Rating 10 per ASTM D610, for rusting; Rating 10 per ASTM D714, for blistering
Pencil Hardness	ASTM D3363	3H
Salt Fog Resistance[3]	ASTM B117, 15,000 hours	Rating 10 per ASTM D610, for rusting; Rating 10 per ASTM D714, for blistering

Meets the requirements of SSPC Paint No. 36, Level 3 for white and light colors. Dark colors may require a clear coat.

Complies with ISO 12944-5 C5I and C5M requirements.

Footnotes:
[1] Finish coat only tested
[2] Primer — Zinc-Clad II Plus
Intermediate — Macropoxy 646
Finish — Acrolon 218 HS
[3] Primer — Zinc-Clad III HS

continued on back

Protective & Marine Coatings

ACROLON™ 218 HS
ACRYLIC POLYURETHANE

Part A	B65-600	Gloss Series
Part A	B65-650	Semi-Gloss Series
Part B	B65V600	Hardener

Revised: July 22, 2021

Product Information

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Recommended Systems

		Dry Film Thickness / ct. Mils	(Microns)
Steel:			
1 ct.	Macropoxy 646	5.0-10.0	(125-250)
1-2 cts.	Acrolon 218 HS Polyurethane	3.0-6.0	(75-150)
Steel:			
1 ct.	Zinc Clad II Plus	2.0-4.0	(50-100)
1 ct.	Macropoxy 646	3.0-10.0	(75-250)
1-2 cts.	Acrolon 218 HS Polyurethane	3.0-6.0	(75-150)
Steel:			
1 ct.	Zinc Clad IV	3.0-5.0	(75-125)
or	Zinc Clad 4100	3.0-5.0	(75-125)
1 ct.	Macropoxy 646	3.0-10.0	(75-250)
1-2 cts.	Acrolon 218 HS Polyurethane	3.0-6.0	(75-150)
Steel:			
1 ct.	Zinc Clad IV	3.0-5.0	(75-125)
1-2 cts.	Acrolon 218 HS Polyurethane	3.0-6.0	(75-150)
Steel:			
1 ct.	Corothane I-GalvaPac Zinc Primer	3.0-4.0	(75-100)
1-2 cts.	Acrolon 218 HS Polyurethane	3.0-6.0	(75-150)
Steel:			
1 ct.	Epoxy Mastic Aluminum II	6.0	(150)
1-2 cts.	Acrolon 218 HS Polyurethane	3.0-6.0	(75-150)
Steel:			
1 ct.	Recoatable Epoxy Primer	4.0-6.0	(100-150)
1-2 cts.	Acrolon 218 HS Polyurethane	3.0-6.0	(75-150)
Concrete/Masonry:			
1 ct.	Kem Cati-Coat HS Epoxy Filler/Sealer	10.0-20.0	(250-500)
1-2 cts.	Acrolon 218 HS Polyurethane	3.0-6.0	(75-150)
Aluminum/Galvanizing:			
1 ct.	DTM Wash Primer	0.7-1.3	(18-32)
1-2 cts.	Acrolon 218 HS Polyurethane	3.0-6.0	(75-150)

FIRETEX ONLY:
Finish Coat for FIRETEX Hydrocarbon Systems:
1 ct. Acrolon 218 HS Polyurethane*
*Consult FIRETEX PFP Specialist for recommended dft range

The systems listed above are representative of the product's use, other systems may be appropriate.

Disclaimer

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Surface Preparation

Surface must be clean, dry, and in sound condition. Remove all oil, dust, grease, dirt, loose rust, and other foreign material to ensure adequate adhesion.

Refer to product Application Bulletin for detailed surface preparation information.

Minimum recommended surface preparation:
* Iron & Steel: SSPC-SP6/NACE 3, 1-2 mil (25-50 micron) profile
* Galvanizing: SSPC-SP1
* Concrete & Masonry: SSPC-SP13/NACE 6, or ICRI No. 310.2R, CSP 1-3
* Primer required

Surface Preparation Standards

	Condition of Surface	ISO 8501-1 BS7079:A1	Swedish Std. SIS055900	SSPC	NACE
White Metal		Sa 3	Sa 3	SP 5	1
Near White Metal		Sa 2.5	Sa 2.5	SP 10	2
Commercial Blast		Sa 2	Sa 2	SP 6	3
Brush-Off Blast		Sa 1	Sa 1	SP 7	4
Hand Tool Cleaning	Rusted	C St 2	C St 2	SP 2	-
	Pitted & Rusted	D St 2	D St 2	SP 2	-
Power Tool Cleaning	Rusted	C St 3	C St 3	SP 3	-
	Pitted & Rusted	D St 3	D St 3	SP 3	-

Tinting

Tint Part A with Maxitoner Colorants.
- Extra white tints at 100% tint strength
- Ultradeep base tints at 150% tint strength

Five minutes minimum mixing on a mechanical shaker is required for complete mixing of color.

Application Conditions

Temperature: 35°F (1.7°C) minimum, 120°F (49°C) maximum (air and surface)
40°F (4.5°C) minimum, 120°F (49°C) maximum (material)
At least 5°F (2.8°C) above dew point

Relative humidity: 85% maximum

Refer to product Application Bulletin for detailed application information.

Ordering Information

Packaging:	1 gallon (3.78L) mix:	5 gallon (18.9L) mix:
Part A:	.86 gal (3.25L)	4.29 gal (16.2L)
Part B:	.14 gal (0.53L)	0.71 gal (2.7L)
(premeasured components)		

Weight: 11.2 ± 0.2 lb/gal ; 1.3 Kg/L mixed, may vary with color

Safety Precautions

Refer to the SDS sheet before use.

Published technical data and instructions are subject to change without notice. Contact your Sherwin-Williams representative for additional technical data and instructions.

Warranty

The Sherwin-Williams Company warrants our products to be free of manufacturing defects in accord with applicable Sherwin-Williams quality control procedures. Liability for products proven defective, if any, is limited to replacement of the defective product or the refund of the purchase price paid for the defective product as determined by Sherwin-Williams. NO OTHER WARRANTY OR GUARANTEE OF ANY KIND IS MADE BY SHERWIN-WILLIAMS, EXPRESSED OR IMPLIED, STATUTORY, BY OPERATION OF LAW OR OTHERWISE, INCLUDING MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Protective & Marine Coatings

SHERWIN WILLIAMS.

ACROLON™ 218 HS
ACRYLIC POLYURETHANE

Part A	**B65-600**	**Gloss Series**
Part A	**B65-650**	**Semi-Gloss Series**
Part B	**B65V600**	**Hardener**

Revised: July 22, 2021

APPLICATION BULLETIN

SURFACE PREPARATIONS

Surface must be clean, dry, and in sound condition. Remove all oil, dust, grease, dirt, loose rust, and other foreign material to ensure adequate adhesion.

Iron & Steel
Remove all oil and grease from surface by Solvent Cleaning per SSPC-SP1. Minimum surface preparation is Commercial Blast Cleaning per SSPC-SP6/NACE 3. For better performance, use Near White Metal Blast Cleaning per SSPC-SP10/NACE 2. Blast clean all surfaces using a sharp, angular abrasive for optimum surface profile (1-2 mils / 25-50 microns). Prime any bare steel the same day as it is cleaned or before flash rusting occurs.

Aluminum
Remove all oil, grease, dirt, oxide and other foreign material by Solvent Cleaning per SSPC-SP1. Primer required.

Galvanized Steel
Allow to weather a minimum of six months prior to coating. Solvent Clean per SSPC-SP1. When weathering is not possible, or the surface has been treated with chromates or silicates, first Solvent Clean per SSPC-SP1 and apply a test patch. Allow paint to dry at least one week before testing adhesion. If adhesion is poor, brush blasting per SSPC-SP7 is necessary to remove these treatments. Rusty galvanizing requires a minimum of Hand Tool Cleaning per SSPC-SP2, prime the area the same day as cleaned or before flash rusting occurs. Primer required.

Concrete and Masonry
For surface preparation, refer to SSPC-SP13/NACE 6, or ICRI No. 310.2R, CSP 1-3. Surfaces should be thoroughly clean and dry. Concrete and mortar must be cured at least 28 days @ 75°F (24°C). Remove all loose mortar and foreign material. Surface must be free of laitance, concrete dust, dirt, form release agents, moisture curing membranes, loose cement and hardeners. Fill bug holes, air pockets and other voids with Steel-Seam FT910. Primer required.

Follow the standard methods listed below when applicable:
ASTM D4258 Standard Practice for Cleaning Concrete.
ASTM D4259 Standard Practice for Abrading Concrete.
ASTM D4260 Standard Practice for Etching Concrete.
ASTM F1869 Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete.
SSPC-SP 13/Nace 6 Surface Preparation of Concrete.
ICRI No. 310.2R Concrete Surface Preparation.

Surface Preparation Standards

	Condition of Surface	ISO 8501-1 BS7079:A1	Swedish Std. SIS055900	SSPC	NACE
White Metal		Sa 3	Sa 3	SP 5	1
Near White Metal		Sa 2.5	Sa 2.5	SP 10	2
Commercial Blast		Sa 2	Sa 2	SP 6	3
Brush-Off Blast		Sa 1	Sa 1	SP 7	4
Hand Tool Cleaning	Rusted	C St 2	C St 2	SP 2	-
	Pitted & Rusted	D St 2	D St 2	SP 2	-
Power Tool Cleaning	Rusted	C St 3	C St 3	SP 3	-
	Pitted & Rusted	D St 3	D St 3	SP 3	-

APPLICATION CONDITIONS

Temperature: 35°F (1.7°C) minimum, 120°F (49°C) maximum (air and surface)
40°F (4.5°C) minimum, 120°F (49°C) maximum (material)
At least 5°F (2.8°C) above dew point

Relative humidity: 85% maximum

APPLICATION EQUIPMENT

The following is a guide. Changes in pressures and tip sizes may be needed for proper spray characteristics. Always purge spray equipment before use with listed reducer. Any reduction must be compliant with existing VOC regulations and compatible with the existing environmental and application conditions.

Reducer/Clean Up:
Spray..............................Reducer R7K15, MEK, Reducer #58, or R7K111
Brush/Roll ....Reducer #132, R7K132, Reducer #58, or R7K111
If reducer is used, reduce at time of catalyzation.

Airless Spray
Pressure.........................2500 - 2800 psi
Hose...............................3/8" ID
Tip ....013" - .017"
Filter ....60 mesh
Reduction.......................As needed up to 10% by volume with R7K15 or R7K111, or up to 9% with MEK, R6K10*

Conventional Spray
Gun ....Binks 95
Cap 63P
Atomization Pressure.....50 - 70 psi
Fluid Pressure................20 - 25 psi
Reduction.......................As needed up to 10% by volume with R7K15 or R7K111, or up to 9% with MEK, R6K10*

Brush
Brush..............................Natural Bristle
Reduction.......................As needed up to 10% by volume*

Roller
Cover ....3/8" woven with solvent resistant core
Reduction.......................As needed up to 10% by volume*

If specific application equipment is not listed above, equivalent equipment may be substituted.

* Note: Reducing more than maximum recommended level will result in VOC exceeding 340g/L

continued on back

Protective & Marine Coatings

SHERWIN WILLIAMS.

ACROLON™ 218 HS
ACRYLIC POLYURETHANE

PART A	B65-600	GLOSS SERIES
PART A	B65-650	SEMI-GLOSS SERIES
PART B	B65V600	HARDENER

Revised: July 22, 2021

APPLICATION BULLETIN

APPLICATION PROCEDURES

Surface preparation must be completed as indicated.

Mix contents of each component thoroughly with low speed power agitation. Make certain no pigment remains on the bottom of the can. Then combine six parts by volume of Part A with one part by volume of Part B (premeasured components). Thoroughly agitate the mixture with power agitation. Re-stir before using.

If reducer is used, add only after both components have been thoroughly mixed.

Apply paint at the recommended film thickness and spreading rate as indicated below:

Recommended Spreading Rate per coat:

	Minimum	Maximum
Wet mils (microns)	**4.5** (112.5)	**9.0** (225)
Dry mils (microns)	**3.0** (75)	**6.0** (150)
~Coverage sq ft/gal (m^2/L)	**175** (4.3)	**346** (8.5)
Theoretical coverage **sq ft/gal** (m^2/L) @ 1 mil / 25 microns dft	**1040** (25.5)	

NOTE: Brush or roll application may require multiple coats to achieve maximum film thickness and uniformity of appearance.

Drying Schedule @ 6.0 mils wet (150 microns):

	@ 35°F/1.7°C	@ 77°F/25°C *50% RH*	@ 120°F/49°C
To touch:	4 hours	1 hour	20 minutes
To handle:	18 hours	9 hours	4 hours
To recoat:			
minimum:	18 hours	8 hours	6 hours
maximum:	3 months	3 months	3 months
To cure:	14 days	7 days	5 days
Pot Life: (reduced 5% with Reducer R7K15)	4 hours	2 hours	45 minutes
Sweat-in-Time:		None	

Drying time is temperature, humidity, and film thickness dependent.
Paint temperature must be at least 40°F (4.5°C) minimum.

Application of coating above maximum or below minimum recommended spreading rate may adversely affect coating performance.

CLEAN UP INSTRUCTIONS

Clean spills and spatters immediately with Reducer #132, R7K132. Clean tools immediately after use with Reducer #132, R7K132. Follow manufacturer's safety recommendations when using any solvent.

DISCLAIMER

The information and recommendations set forth in this Product Data Sheet are based upon tests conducted by or on behalf of The Sherwin-Williams Company. Such information and recommendations set forth herein are subject to change and pertain to the product offered at the time of publication. Consult your Sherwin-Williams representative to obtain the most recent Product Data Information and Application Bulletin.

PERFORMANCE TIPS

Stripe coat all crevices, welds, and sharp angles to prevent early failure in these areas.

When using spray application, use a 50% overlap with each pass of the gun to avoid holidays, bare areas, and pinholes. If necessary, cross spray at a right angle.

Spreading rates are calculated on volume solids and do not include an application loss factor due to surface profile, roughness or porosity of the surface, skill and technique of the applicator, method of application, various surface irregularities, material lost during mixing, spillage, overthinning, climatic conditions, and excessive film build.

Excessive reduction of material can affect film build, appearance, and adhesion.

Do not apply the material beyond recommended pot life.

Do not mix previously catalyzed material with new.

In order to avoid blockage of spray equipment, clean equipment before use or before periods of extended downtime with Reducer #15, R7K15 or MEK, R6K10.

Mixed coating is sensitive to water. Use water traps in all air lines. Moisture contact can reduce pot life and affect gloss and color.

Quick-Thane Urethane Accelerator is acceptable for use. See data page 5.97 for details.

E-Z Roll Urethane Defoamer is acceptable for use. See data page 5.99 for details.

If maximum recoat time is exceeded, a light abrasion may be necessary to roughen the surface to promote adhesion before recoating.

Refer to Product Information sheet for additional performance characteristics and properties.

SAFETY PRECAUTIONS

Refer to the SDS sheet before use.

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WARRANTY

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WPE HEATX HEAT TRANSFER AND HYDRAULIC HEAD LOSS PROGRAM **HEATX 01.XLS - REV 21 - 6/2/2023**

HILO WWTP
HILO, HI
X250480

DATE: 05/29/25
BY: JDG

SLUDGE SPECIFIC GRAVITY	1.000
SLUDGE VISCOSITY	1.00 cSt
SLUDGE SPECIFIC HEAT	1.00 BTU/(LBm-°F)

THIS IS A SLUDGE HEATING APPLICATION

PROCESS HEAT TRANSFER REQUIRED	425,000 BTU / HR
EXCHANGER WATER TUBE MATERIAL	SA 53 E/S ALLOY STEEL
EXCHANGER WATER SIDE TUBE	8 INCH SCH 40 PIPE
EXCHANGER WATER SIDE TUBE WALL THICKNESS	0.322 IN
EXCHANGER SLUDGE TUBE MATERIAL	SA 312 E/S STAINLESS STEEL
EXCHANGER SLUDGE SIDE TUBE	6 INCH SCH 10 PIPE
EXCHANGER SLUDGE SIDE TUBE WALL THICKNESS	0.134 IN
EXCHANGER FRAME LENGTH	6.00 FT
NUMBER OF TUBES PER PASS	6
NUMBER OF PASSES - WATER	1
NUMBER OF PASSES - SLUDGE	1
HEAT TRANSFER AREA - SLUDGE PIPE O.D. AREA	70.9 FT²
HEAT TRANSFER AREA - SLUDGE PIPE I.D. AREA	68.0 FT²
WATER FLOW PER PASS	100 GPM
WATER INLET TEMPERATURE	150.0 °F
SLUDGE FLOW PER PASS	160 GPM
SLUDGE INLET TEMPERATURE	98.0 °F
THEORETICAL PRESSURE DROP WATER SIDE	1.7 FT WC
THEORETICAL PRESSURE DROP SLUDGE SIDE	0.3 FT WC
OVERALL HEAT TRANSFER COEFFICIENT WITHOUT FOULING	163.6 BTU/(HR-FT²-°F)
WATER OUTLET TEMPERATURE - NO FOULING	139.7 °F
WATER OUTLET TEMPERATURE DIFFERENCE - NO FOULING	10.3 °F
SLUDGE OUTLET TEMPERATURE - NO FOULING	104.4 °F
HEAT TRANSFERRED - NO FOULING	506,181 BTU/HR
APPLIED SLUDGE HEAT TRANSFER FOULING FACTOR	0.0010 (HR-FT²-°F)/BTU
OVERALL HEAT TRANSFER COEFFICIENT WITH FOULING FACTOR	139.8 BTU/(HR-FT²-°F)
WATER OUTLET TEMPERATURE WITH FOULING	141.0 °F
WATER OUTLET TEMPERATURE DIFFERENCE WITH FOULING	9.0 °F
SLUDGE OUTLET TEMPERATURE WITH FOULING	103.6 °F
HEAT TRANSFERRED WITH FOULING	442,780 BTU/HR

STAMPED ANCHORAGE

DESIGN CALCULATIONS

TO BE PROVIDED UNDER A SEPARATE COVER WHEN AVAILABLE

Walker Process Equipment
Partial Installation List
Heat Exchangers - Standalone

Contract	Location	Qty.	Size	Year	Engineer
M30310	WINSTON-SALEM, NC	3	1000 E EXCHANGER	1994	HAZEN & SAWYER
M40600	ELDRIDGE, IA	1	1000 E EXCHANGER	1995	SHIVE, HATTERY & ASSOCIATES
M70101	MIDLAND, TX	2	1000 E EXCHANGER	1997	HDR ENGINEERING INC.
M70110	DALLAS, TX	3	1000 E EXCHANGER	1997	HDR ENGINEERING INC.
M70562	DURHAM, NC	2	1000 E EXCHANGER	1998	HAZEN & SAWYER
M70571	DURHAM, NC	2	1000 E EXCHANGER	1998	HAZEN & SAWYER
P10080	LIBERTYVILLE, IL	1	1000 E EXCHANGER	2001	REZEK, HENRY, MEISEN. & GENDE
P10210	YONKERS, NY	2	1000 E EXCHANGER	2001	SAVIN ENGINEERING
P60230	COLUMBIA, MO	1	1000 E EXCHANGER	2005	NONE
P70270	COLUMBIA, MO	1	1000 E EXCHANGER	2007	CITY
P70620	MARQUETTE, MI	2	1000 E EXCHANGER	2007	DONAHUE & ASSOCIATES
P90353	FORT DODGE, IA	1	1000 E EXCHANGER	2009	MCCLURE ENGINEERING
Q20241	VIRGINIA, MN	1	1000 E EXCHANGER	2011	SEH ENGINEERS
Q30220	SAN FRANCISCO, CA	2	1000 E EXCHANGER	2012	CH2M HILL
Q30420	FORT DODGE, IA	1	1000 E EXCHANGER	2012	McCLURE ENGINEERING
Q50653	WEST PALM BEACH, FL	4	1000 E EXCHANGER	2015	HAZEN & SAWYER
P50590	DELPHOS, OH	1	1040-3090 E EXCHANGER	2005	UNKNOWN
M10623	ORR, MN	1	110 E EXCHANGER	1992	JOHN BAKER ENGINEERING
Q00800R	MARATHON, WI	1	110 E EXCHANGER	2010	NONE
X210440	BENSON, MN	1	140 E	2021	STANTEC
Q90061	MIAMI-DADE, FL (SOUTH)	8	1400 E	2018	STANTEC
M20406	HOUGHTON-HANCOCK,MI	1	1500 E EXCHANGER	1993	U.P. ENGINEERING / TDKA
M50350	OCEAN CO., NJ	1	1500 E EXCHANGER	1995	METCALF & EDDY, INC.
M80201	NEENAH-MENASHA, WI	1	1500 E EXCHANGER	1998	McMAHON ASSOCIATES
P10621	JACKSON, MI	2	1500 E EXCHANGER	2001	EARTH TECH.
P20760	EAST PROVIDENCE, RI	3	1500 E EXCHANGER	2002	CAMP, DRESSER & MCKEE
P30810	PELHAM, SC	3	1500 E EXCHANGER	2003	BLACK & VEATCH
P50290	ROANOKE, VA	1	1500 E EXCHANGER	2004	HAZEN & SAWYER

Walker Process Equipment
Partial Installation List
Heat Exchangers - Standalone

Contract	Location	Qty.	Size	Year	Engineer
P50291	ROANOKE, VA	1	1500 E EXCHANGER	2005	HAZEN & SAWYER
P70592	JOLIET, IL	2	1500 E EXCHANGER	2007	CLARK, DIETZ & ASSOCIATES
P80820	ABERDEEN, SD	1	1500 E EXCHANGER	2008	BANNER CONSULTING ENGINEERS
Q20690	RICHLAND CENTER, WI	1	1500 E EXCHANGER	2012	PROBST GROUP
X210013	WAUSAU, WI	1	1750 E	2020	DONOHUE & ASSOC
X230411	AUBURN, NY	1	1750 E	2023	BROWN & CALDWELL
X240270	HONOULIULI, HI	3	1900 E	2023	BGE
P40940	YORKVILLE, IL	1	200 E EXCHANGER	2004	WALTER DEUCHLER & ASSOCIATES
X210250	GREENVILLE, SC	2	2000 E	2020	OWNER
X210572	LA CROSSE, WI	4	2000 E	2021	DONOHUE & ASSOC.
X210700	GREENVILLE, SC	1	2000 E	2021	HAZEN & SAWYER
M10852	TULSA, OK	3	2000 E EXCHANGER	1992	BLACK & VEATCH
M30273	AMARILLO, TX	3	2000 E EXCHANGER	1994	BLACK & VEATCH
M40200	SYVESTER WHEY, WI	1	2000 E EXCHANGER	1994	APPLIED TECHNOLOGY
M50420	EAST ROCKAWAY, NY	5	2000 E EXCHANGER	1995	HAZEN & SAWYER
M70882	LOUISVILLE, KY	1	2000 E EXCHANGER	1998	PRO CORP
M80202	NEENAH-MENASHA, WI	1	2000 E EXCHANGER	1998	McMAHON ASSOCIATES
M80580	BAKERSFIELD, CA	1	2000 E EXCHANGER	1999	BOYLE ENGINEERING CORP
M90472	GREENVILLE, SC	2	2000 E EXCHANGER	1999	HAZEN & SAWYER
P30190	ALEXANDRIA, VA	4	2000 E EXCHANGER	2002	CH2M HILL
P30563	WAUPUN, WI	1	2000 E EXCHANGER	2003	EARTH TECH.
Q30430	CHARLOTTSVILLE, VA	3	2000 E EXCHANGER	2013	HAZEN & SAWYER
Q30480	WINSTON-SALEM, NC	1	2000 E EXCHANGER	2013	WGR
Q30733	EAU CLAIRE, WI	3	2000 E EXCHANGER	2013	DONAHUE & ASSOCIATES
Q30830	IRVINE, CA	3	2000 E EXCHANGER	2013	BLACK & VEATCH
Q50070	WINCHESTER, VA	3	2000 E EXCHANGER	2014	BLACK & VEATCH
Q50654	WEST PALM BEACH, FL	1	2000 E EXCHANGER	2015	HAZEN & SAWYER
Q60662	ST. PETERSBURG, FL	2	2000 E EXCHANGER	2016	BROWN & CALDWELL

Walker Process Equipment
Partial Installation List
Heat Exchangers - Standalone

Contract	Location	Qty.	Size	Year	Engineer
M50730	BRITT, IA	1	250 E EXCHANGER	1996	WALLACE, HOLLAND, KASTLER, S
P20571	HANOVER PK: CHICAGO, IL	1	250 E EXCHANGER	2002	MWRD OF GREATER CHICAGO
P41040	BRUNSWICK CO, NC	1	250 E EXCHANGER	2004	NONE
X250320	NEW YORK, NY - TALLMAN ISL	2	2500 (EAST SIDE)	2025	NY DEP
X250500	SHEBOYGAN, WI	2	2500 E	2025	DONOHUE
X250140	NEW YORK, NY - TALLMAN ISL	2	2500 E	2024	DEP
X220660	BELOIT, WI	2	2500 E	2022	DONOHUE & ASSOC.
X240140	DAYTON, OH	6	2500 E	2023	HAZEN & SAWYER
M80484	GWINNETT CO., GA	2	2500 E EXCHANGER	1999	MULTIPLE . . .
P10741	GVL: HONG KONG	2	2500 E EXCHANGER	2001	EARTH TECH.
P30652	ELGIN, IL	1	2500 E EXCHANGER	2003	BAXTER AND WOODMAN
P30820	GWINNETT CO, GA	3	2500 E EXCHANGER	2003	JORDAN, JONES & GOULDING
P40781	INLAND EMPIRE RP#2, CA	2	2500 E EXCHANGER	2004	OWNER
P50160R	SAN ANTONIO, TX	2	2500 E EXCHANGER	2004	CITY
P50420	AVON, CO	1	2500 E EXCHANGER	2005	BROWN & CALDWELL
P50790	PAPILLION CREEK, OMAHA, NE	6	2500 E EXCHANGER	2005	HDR ENGINEERING INC.
P60150	HEART OF THE VALLEY, WI	2	2500 E EXCHANGER	2005	McMAHON ASSOCIATES
P60380	DURHAM, ONTARIO	2	2500 E EXCHANGER	2005	CH2M HILL
P60701	MICHIGAN STATE UNIV.	1	2500 E EXCHANGER	2006	MICHIGAN STATE UNIVERSITY
P80140R	SAN ANTONIO, TX	1	2500 E EXCHANGER	2007	NONE
Q00810R	SAN ANTONIO, TX	2	2500 E EXCHANGER	2010	CITY ENGINEERS
Q20560R	SAN ANTONIO, TX	1	2500 E EXCHANGER	2012	CITY ENGINEERS
Q40100R	SAN ANTONIO, TX	2	2500 E EXCHANGER	2013	CITY ENGINEERS
Q50340	OAKVILLE, ONTARIO	1	2500 E EXCHANGER	2014	STANTEC CONSULTING, LTD.
P50680	KINGSTON, ONTARIO	1	2500 ESS EXCHANGER	2005	J. L. RICHARDS
Q50651	WEST PALM BEACH, FL	3	2500 ESS EXCHANGER	2015	HAZEN & SAWYER
X250420	TAMPA, FL	2	2600 E	2025	HAZEN & SAWYER
X230530	TAMPA, FL	1	2600 E	2023	HAZEN & SAWYER

Walker Process Equipment
Partial Installation List
Heat Exchangers - Standalone

Contract	Location	Qty.	Size	Year	Engineer
X230540	LITTLE FERRY, NJ	5	2600 E	2023	AECOM
X240580	RICHMOND, VA	4	2700 E	2024	GREELEY & HANSEN
X240080	NEW YORK, NY	4	2750 E	2023	CDM SMITH
X230100	OMAHA, NE	2	3000 E	2022	HDR
X230140	JEROME, ID - MONTAUK	1	3000 E	2022	AECOM
X230210	SUFFOLK, VA	2	3000 E	2022	HAZEN & SAWYER
M20150	NATL.INST.HEALTH -MD	1	3000 E EXCHANGER	1993	EA MUELLER
M20360	BOSTON, MA	10	3000 E EXCHANGER	1993	BLACK & VEATCH
M40410	BOSTON, MA	8	3000 E EXCHANGER	1995	CAMP, DRESSER & MCKEE
M80390	MEDELLIN, COLUMBIA	2	3000 E EXCHANGER	1999	GREELEY & HANSEN
P00361	ESSEX & UNION JM, NJ	2	3000 E EXCHANGER	2000	CME ASSOCIATES
P00610	CHICAGO, IL - EGAN	3	3000 E EXCHANGER	2001	BLACK & VEATCH
P10400	ROCKFORD, IL	4	3000 E EXCHANGER	2001	BLACK & VEATCH
P10530	JACKSONVILLE, FL	2	3000 E EXCHANGER	2001	BLACK & VEATCH
P10900	HENRICO CO, VA	4	3000 E EXCHANGER	2001	HAZEN & SAWYER
P30520	RACINE, WI	1	3000 E EXCHANGER	2003	EARTH TECH.
P30892	DELAFIELD, WI	2	3000 E EXCHANGER	2003	FOTH & VAN DYKE
P30900	PHILADELPHIA, PA	12	3000 E EXCHANGER	2003	VINOKUR & PACE ENG SERVICES
Q10060	TALLAHASSEE, FL	2	3000 E EXCHANGER	2010	HAZEN & SAWYER
Q10622	ST LOUIS, MO	1	3000 E EXCHANGER	2011	BLACK & VEATCH
Q60661	ST. PETERSBURG, FL	3	3000 E EXCHANGER	2016	BROWN & CALDWELL
Q70140	SPRINGFIELD, MO	2	3000 E SS	2016	BLACK & VEATCH
X220350	HOLLAND, MI	2	3000 ESS	2021	NONE
P40030	ELKHART, IN	1	3000 ESS EXCHANGER	2003	MALCOLM PIRNIE
P60702	MICHIGAN STATE UNIV.	1	3000 ESS EXCHANGER	2006	MICHIGAN STATE UNIVERSITY
X230372	OMAHA, NE	2	3057 E (COOLING)	2023	HDR
X210610	DES MOINES, IA	4	3200 E	2021	CITY
Q30400	DES MOINES, IA	2	3200 E EXCHANGER	2012	CDM ENGINEERS

Walker Process Equipment
Partial Installation List
Heat Exchangers - Standalone

Contract	Location	Qty.	Size	Year	Engineer
P60560	TORONTO, ONTARIO	8	3400 E EXCHANGER	2006	CH2M HILL
Q70042	SAN JOSE-SANTA CLARA, CA	4	3400 E EXCHANGER	2016	SUBURBAN CONSULTING ENGRS
P10140	NEW YORK, NY	8	3500 E EXCHANGER	2001	MALCOLM PIRNIE / HAZEN & SAWYER
Q50652	WEST PALM BEACH, FL	2	3700 E EXCHANGER	2015	HAZEN & SAWYER
X210510	MENOMINEE, MI	1	375 E	2021	CITY
X230430	SALEM, OH	1	400 E	2023	BURGESS & NIPLE
M60460	ATHENS, PA	1	400 E EXCHANGER	1997	?
P40640	MOREHEAD, KY	1	400 E EXCHANGER	2004	THERMAL PROCESS SYSTEMS
P70090	SHELAND FARMS: ADAMS, NY	1	400 E EXCHANGER	2006	STEARNS & WHELER
P80520	ORVILLE, OH	1	400 E EXCHANGER	2008	BURGESS & NIPLE LTD
Q10030	L'ANSE, MI	1	400 E EXCHANGER	2010	UP ENGINEER
Q30150	HUDSON, WI	1	400 E EXCHANGER	2012	TKDA
P70140	LAKELAND, FL	1	4000 E EXCHANGER	2006	CHICAGO BRIDGE & IRON
Q30101	KANKAKEE, IL	2	4000 E EXCHANGER	2012	STRAND & ASSOC.
P80354	AURORA, IL (FOX METRO)	1	4000 ESS EXCHANGER	2007	WALTER DEUCHLER & ASSOCIATES
X250480	HILO, HI	2	425 E(S)	2025	CAROLLO
X220180	DESAEGHER DAIRY	2	4700 E	2021	BROWN & CALDWELL
X220780	DECLO, ID	6	4710 E	2022	BROWN & CALDWELL
M30680	GOSHEN, IN	2	500 E EXCHANGER	1994	GREELEY & HANSEN
M70160	STURGEON BAY, WI	1	500 E EXCHANGER	1997	McMAHON ASSOCIATES
P10622	JACKSON, MI	1	500 E EXCHANGER	2001	EARTH TECH.
P40860	BOWLING GREEN, OH	1	500 E EXCHANGER	2004	NONE
P60030R	ASTON, PA	1	500 E EXCHANGER	2005	NONE
P60300	WESTMINSTER, CO	1	500 E EXCHANGER	2005	CAMP, DRESSER & MCKEE
P70600	LODI, CA	1	500 E EXCHANGER	2007	WEST YOST
Q60270	HILLSDALE, MI	1	500 E EXCHANGER	2015	FLEIS & VANDERBRINK
Q60621	ROCKY RIVER, OH	2	500 E EXCHANGER	2016	AECOM
M70561	DURHAM, NC	4	500 E EXCHANGER	1998	HAZEN & SAWYER

Walker Process Equipment
Partial Installation List
Heat Exchangers - Standalone

Contract	Location	Qty.	Size	Year	Engineer
X230340	AURORA, IL (FOX METRO)	2	5000 E	2023	DISTRICT
M30672	CALUMET, IL	4	5000 E EXCHANGER	1994	MWRD OF GREATER CHICAGO
P80355	AURORA, IL (FOX METRO)	1	5000 E EXCHANGER	2007	WALTER DEUCHLER & ASSOCIATES
Q10480	AURORA, IL - FOX METRO	1	5000 E EXCHANGER	2011	WALTER DEUCHLER & ASSOCIATES
X240620	KENT, OH	1	550 E	2024	CTI ENGINEERS, INC
Q00550	RHINELANDER, WI	1	550 ESS EXCHANGER	2010	TOWN & COUNTRY ENGINEERS
Q70041	SAN JOSE-SANTA CLARA, CA	8	5800 E EXCHANGER	2016	SUBURBAN CONSULTING ENGRS
X210580	JOHNSON CO, KS	1	600 E	2021	CDM SMITH
P90532	JOHNSON CO, KS	2	600 E EXCHANGER (FOG)	2009	HDR / ARCHER
P40200	INLAND EMPIRE UTIL., CA	1	6000 E EXCHANGER	2003	OWNER
Q50060	CHICAGO, IL (STICKNEY)	6	6000 E EXCHANGER	2014	MWRD OF GREATER CHICAGO
Q50320	LONGVIEW, TX	3	675 E EXCHANGER	2014	GARVER ENGINEERS
X240660	MISSION VIEJO, CA	2	700 E	2024	VISKASE
M30140	EIELSON AFB, AK	2	700 E EXCHANGER	1994	ARMY CORP. OF ENGINEERS
M80082	ELKO, NV	1	700 E EXCHANGER	1998	GREELEY & HANSEN
M80630	NORWALK, WI	1	700 E EXCHANGER	1999	---
P10482	WILLMAR, MN	2	700 E EXCHANGER	2001	BONESTRO, ROSENE & ANDERLIK
P20310	LINCOLN, NE	2	700 E EXCHANGER	2002	HDR ENGINEERING INC.
P31130	CHIPPEWA FALLS, WI	1	700 E EXCHANGER	2003	EARTH TECH.
P70750	GERVAIS FAMILY FARM: VT	1	700 E EXCHANGER	2007	STAN WEEKS
Q00582	JANESVILLE, WI	1	726 ESS EXCHANGER	2010	AECOM
X240040	GOLETA, CA	1	750 E	2023	HAZEN & SAWYER
M10694	COLUMBIA, MO	1	750 E EXCHANGER	1992	METCALF & EDDY, INC.
M90500	DELTA CHART. TWP, MI	1	750 E EXCHANGER	1999	---
P00601	FORT COLLINS, CO	1	750 E EXCHANGER	2001	MONTGOMERY WATSON HARZA
P30562	WAUPUN, WI	1	750 E EXCHANGER	2003	EARTH TECH.
P90531	JOHNSON CO, KS	1	750 E EXCHANGER	2009	HDR / ARCHER
Q50570	ALAXENDRIA, MN	2	750 E EXCHANGER	2015	BROWN & CALDWELL

Walker Process Equipment
Partial Installation List
Heat Exchangers - Standalone

Contract	Location	Qty.	Size	Year	Engineer
X220470	NEW YORK, NY	4	7700 E	2022	NEW YORK CITY EPB
X230371	OMAHA, NE	8	8296 E (THERMOPHILIC)	2023	HDR
X200375	SIMI VALLEY, CA	2	900 E	2020	SCHNEIDER ELECTRIC
Q70670	ST LOUIS, MO	2	E 1000	2017	OWNER
Q80632	MIDLAND, TX	2	E 1000	2018	HDR
Q90160	ST LOUIS, MO	2	E 1100	2018	CITY ENGINEERS
Q90062	MIAMI-DADE, FL (SOUTH)	4	E 1400	2018	STANTEC
Q70540	HUTCHINSON, KS	2	E 1500	2017	CDM SMITH
Q80480	LOVELAND, CO	2	E 1580	2018	BROWN & CALDWELL
Q80080	ST CLOUD, MN	2	E 1750	2017	DONOHUE & ASSOCIATES
Q70330	MIAMI-DADE CO, FL	4	E 2000	2016	MWH GLOBAL
Q70513	MANITOWOC, WI	2	E 2000	2017	STRAND & ASSOC.
Q70591	LAKELAND, FL	1	E 2000	2017	HDR ENGINEERING INC.
Q80690	MIAMI-DADE, FL (CENTRAL)	4	E 2000	2018	STANTEC
Q90570	SIMPSONVILLE, SC	1	E 2000	2019	HAZEN & SAWYER
Q80460	OMAHA, NE	1	E 2300	2018	HDR
X200010	MARICOPA, AZ	2	E 3380	2019	BROWN & CALDWELL
Q70480	FORT WAYNE, IN	2	E 3600	2017	DONOHUE & ASSOC
X200470	CHICAGO, IL	12	E 5000	2020	CHICAGO MWRD
M00132	LEXINGTON, NE	1	E EXCHANGER	1991	HDR ENGINEERING INC.
M20652	STICKNEY, IL	12	E EXCHANGER	1993	MWRD OF GREATER CHICAGO
M50650	MOBILE, AL	2	E EXCHANGER	1996	BCM ENGINEERS
Q70592	LAKELAND, FL	1	E(S) 5000	2017	HDR ENGINEERING INC.

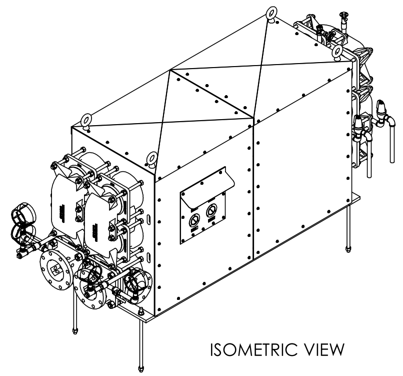
CONTRACT DRAWINGS

NOTES:
1. MAXIMUM OPERATING PRESSURE IS 50 PSIG

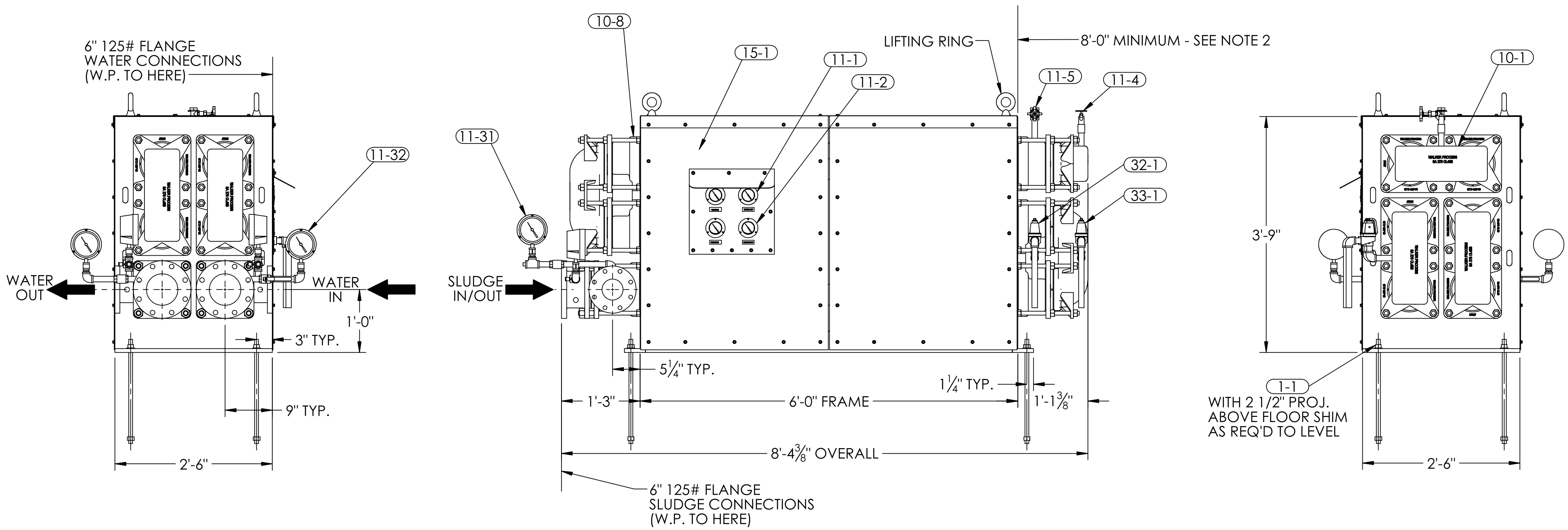
2. THIS CLEARANCE MUST BE ALLOWED ON EITHER ONE OF THE ENDS FOR REMOVAL OF TUBES

3. CONCRETE PAD (IF REQUIRED) MUST BE LEVEL. IF MOUNTING SURFACE IS NOT LEVEL, SHIM OR GROUT UNIT LEVEL (NOT BY W.P.E.)

MARK	DESCRIPTION	SEE DRAWING
01-01	ANCHORS, 3/4" X 19 7/8" W/ NUTS & WASHERS	
10-01	SLUDGE RETURN BEND 6"	4L8698
10-08	WATER RETURN BEND 8"	4L8698
11-01	THERMOMETER - WATER	
11-02	THERMOMETER - SLUDGE	
11-04	SLUDGE TUBE VENT	
11-05	WATER TUBE VENT	
11-31	PRESSURE GAUGE - WATER	
11-32	PRESSURE GAUGE - SLUDGE	
11-41	SLUDGE DRAIN VALVE	
11-42	WATER DRAIN VALVE	
15-01	SIDE PANEL	
32-01	PRESSURE RELIEF VALVE - SLUDGE	
33-01	PRESSURE RELIEF VALVE - WATER	



ISOMETRIC VIEW



HEATX: E-475
NO. UNITS: THREE (3)
SPECIAL ORIENTATION: SLUDGE CASTINGS ON BOTTOM LEFT AND RIGHT WHEN FACING CASTING
SINGLE PASS SLUDGE / SINGLE PASS WATER
6" SLUDGE PIPES, 8" WATER PIPES

REVISIONS		REMARKS
BY / CHKD / DATE	3	
BY / CHKD / DATE	2	
BY / CHKD / DATE	1	

STANDARD TOLERANCE UNLESS OTHERWISE SPECIFIED

Decimals		Fractions
.X	± .1	±
.XX	± .03	Angular
.XXX	± .010	±

	NAME	DATE
DRAWN	JDG	6/17/2025
CHECKED		
ENG APPR.		

FILE REFERENCE

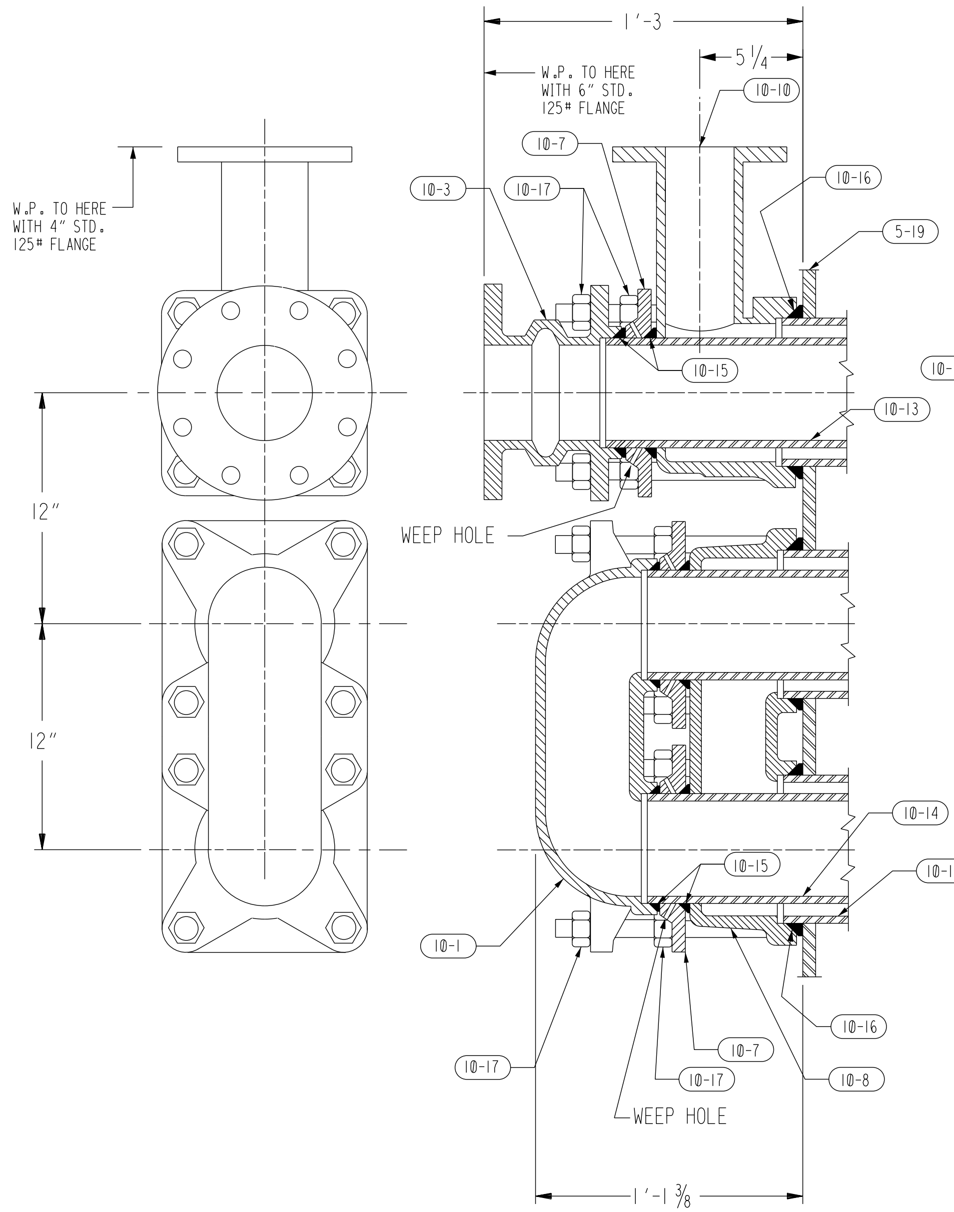
PART NUMBER

SHEET 1 OF 1

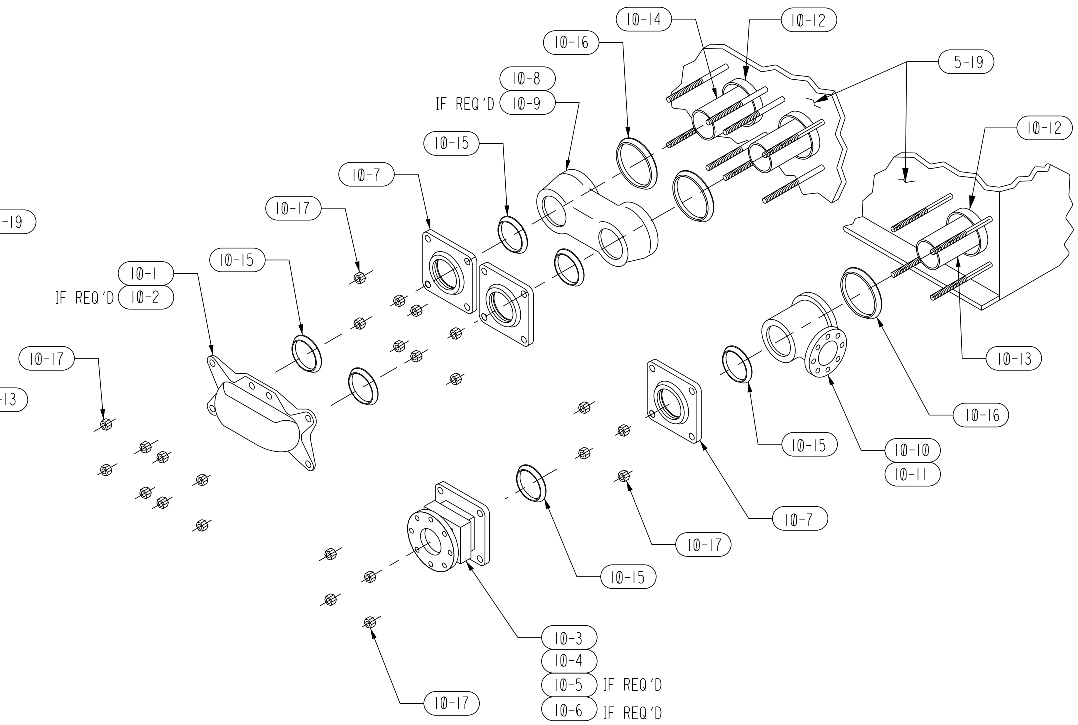
Walker Process Equipment
Division of McNish, LLC
AURORA, ILLINOIS U.S.A.

TITLE: GENERAL ASSEMBLY
TWO (2) E-425 HEATX
HILO, HI

CONTRACT X250480 | DRG 1 | REV 1



TYPICAL SECTION THRU END CASTINGS



A- INTERMEDIATE COLLAR, 10-7, HAS 1/4" WEEP HOLE. ASSEMBLE SO THIS WEEP HOLE IS AT BOTTOM.

B- BEFORE ASSEMBLY COAT ALL GASKETS WITH GREASE TO PREVENT BINDING.

C- SLUDGE TUBES AT ADAPTER ENDS ARE SLIGHTLY LONGER THAN AT RETURN BENDS.

D- DRAW NUTS DOWN TIGHT ON INTERMEDIATE COLLAR BEFORE CONTINUING ASSEMBLY. AFTER COMPLETING ASSEMBLY, AND NUTS ARE TIGHT ON SLUDGE ADAPTERS AND RETURN BENDS. TIGHTEN NUTS ON COLLARS ONLY SLIGHTLY. (26 FT.LB. MAX.)

E- END CASTINGS ARE DESIGNED TO PREVENT CONTAMINATION OF THE BOILER WATER BY ANY MATERIAL CIRCULATING THROUGH THE SLUDGE TUBES. ALL PASSAGEWAYS PASS 4 1/2" DIAMETER SPHERES.

BY / CHKD / DATE	REVISIONS	REMARKS
BY DKJ	DATE 2-3-93	
BY CAP	DATE 21-22-97	CONVERTED TO CAD

The use of anchor types, size, embedment or method other than that shown or provided by Walker Process Corporation will be done at the contractor's risk

This Drawing is the property of Walker Process Equipment and is to be used only in connection with the performance of work by Walker Process. Reproduction in whole or in part for any other purpose is expressly forbidden.

	DATE	BY
DRAWN		
CHECKED		
APPR.		
SCALE		

FILE REFERENCE 4L8698-1

Division McNish Corporation

Walker Process Equipment
Division of McNish Corporation
AURORA, ILLINOIS U.S.A.

END CASTING ASSEMBLY
6" & 8"

CONTRACT | DRG | REV 0

CAD FILE: 4L8698

VENDOR INFORMATION

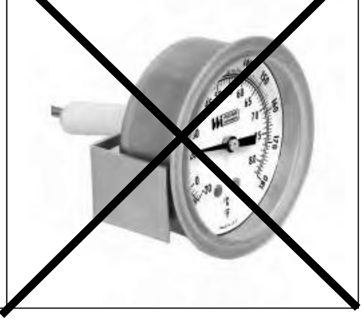
MK 11-1 WATER THERMOMETERS

MK 11-2 SLUDGE THERMOMETERS

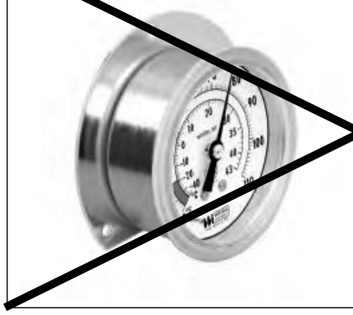
Series 716
Panel Mounting

20FB - Front Flange, Flush Mount

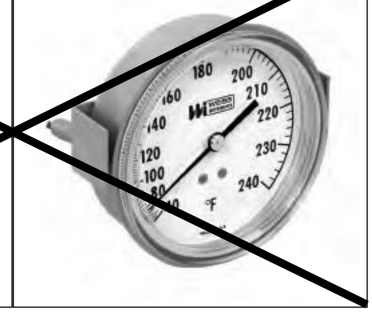
20UB - U-Clamp, Panel Mount



25BB - Back Flange, Wall Mount



35QB - U-Clamp Panel Mount



CODE	CATALOG NO.	DESCRIPTION
I		**CASE SIZE**
	15	1-1/2" Case
	20	2" Case
	25	2-1/2" Case
	35	3-1/2" Case
II		**CASE TYPE**
	F	Front Flange Flush Mount, 2, 2-1/2"
	U	U-Clamp Flush Mount, 1-1/2 ,2, 2-1/2"
	B	Back Flange Surface Mount, 2, 2-1/2"
	S	Front Flange Semi-Flush Mount, 3-1/2"
	J	U-Clamp Semi-Flush Mount, 3-1/2"
	Q	U-Clamp Flush Mount, 3-1/2"
III		**CASE CONNECTION**
	B	Back Connection - Standard
	L	Lower Connection
IV		**CASE FRONT AND WINDOW**
	3	Molded one-piece Lexan
	4	Stainless Steel Ring & Glass
	5	Chrome Plated Brass, Glass
	8	Crimped Ring, S.S., Plastic or Glass - Waterproof
	9	Threaded one-piece Lexan

V — THERMAL SYSTEM

CATALOG NO.	Bulb Type	Bulb Material	Capillary Material	Capillary Protection
1	Plain	Copper	Copper	None
2	Plain	Copper	Copper	Cu. Braid
3	Plain	316 S.S.	316 S.S.	None
4	Union	Copper	Copper	None*
5	Union	Copper	Copper	Cu. Braid
6	Union	316 S.S.	Copper	None
7	Union	316 S.S.	Copper	Cu. Braid
8	Union	316 S.S.	316 S.S.	Intlock Armor*
9	Union	316 S.S.	316 S.S.	None

*At bulb end 31/2" long. All systems have S.S. Spring capillary protection at indicating head. No. 5–7 system include S.S. spring at bulb end – S.S. jam nut standard on No. 8–9 system.

VI — BULB SELECTION

CATALOG NO.	Thermal System No. 4 Thru 9 (Union) Dia.	Thermal System No. 4 Thru 9 (Union) Length	Thermal System No. 1 – 3 (Plain) Dia.	Thermal System No. 1 – 3 (Plain) Length	Max. System Len.
0	7/16	1.87"	N.A.	N.A.	10Ft.
1	7/16	2.5"	N.A.	N.A.	10Ft.
2	7/16	3.4"	3/8	3.4	25Ft.
3	7/16	5.4"	3/8	4.9	50Ft.
4	7/16	7.4"	3/8	7.9	99Ft.
6	N.A.	N.A.	3/8	2.5	05Ft.
7	N.A.	N.A.	5/16	2.5	10Ft.

*Bulb lengths are standard for system lengths shown. Longer standard bulbs can be applied to any shorter system length.The bulb length shown is for comparative purposes, relating system length to bulb length. See Bulb Chart on Page 4 for Dim.

SPECIFICATIONS

CASE & RING - drawn stainless steel case with snap-in "O" ring gasketed clear lexan or glass window.

DIAL - white coated aluminum with black markings.

POINTER - slotted adjustable type which permits "zero set" calibration.

BOURDON TUBE - precision geared brass movement and phosphor bronze tube.

THERMAL SYSTEMS - available in nine standard types, with system lengths from 6 inches thru 100 feet. Materials for the case and ring assembly are as shown in the Coding Charts.

PROCESS CONN. FITTING - separable socket, union or air duct flange connections.

ACCURACY - one scale division throughout range. Built with NSF accuracy and compliance, see our complete listing at www.nsf.org.

CODE	CATALOG NO.	DESCRIPTION
VII		**PROCESS CONNECTION FITTING**
	1	Union Connection 1/2" NPT
	2	Union Connection 3/4" NPT
	3	Separable Socket 1/2" NPT
	4	Separable Socket 3/4" NPT
	7	Air Duct Flange, Union Conn., Aluminum only
	9	Plain bulb, no fitting
VIII		**PROCESS CONNECTION MATERIAL**
	1	Brass
	2	304 Stainless Steel
	3	316 Stainless Steel
	5	Aluminum (Air Duct Flange only)
	0	None – Plain Bulb
IX A-B	See 15 ft	**SYSTEM LENGTH SPECIFYING** 9A - Use 2 digits: EX. 05, 10, 99 for length required. 9B - Use FT for feet, use IN for inches
IX C		**SYSTEM TIN PLATING** (IF REQUIRED) 9C - Use TIN (No. 1, 4 & 5 Thermal System*) NOTE: All other system plating consult factory. *Should be specified for Foodservice Equipment Thermometers

EXAMPLE FOR OBTAINING COMPLETE CATALOG NUMBER FROM CODES - SERIES 716

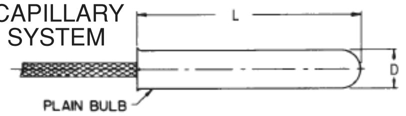
CODE I–IV (HEAD ASSY)		CODE V–VIII (THERMAL SYSTEM)		CODE IX A – B (CAPILLARY LENGTH)
2 0 F B 3	—	1 6 9 0	—	0 5 F T

CODE IX C	RANGE
— — —	40-0-60 F&C

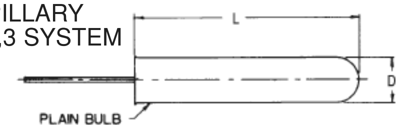
Thermal Systems

Code V

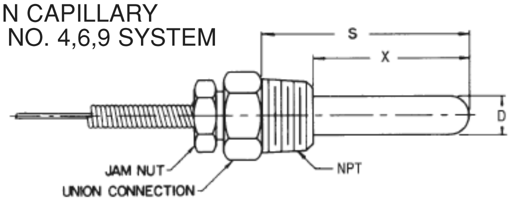
BRAIDED CAPILLARY
CAT. NO. 2 SYSTEM



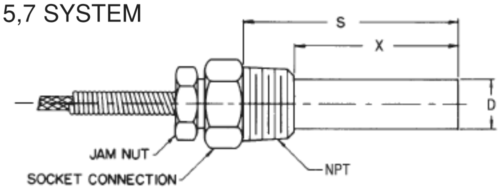
PLAIN CAPILLARY
CAT. NO. 1,3 SYSTEM



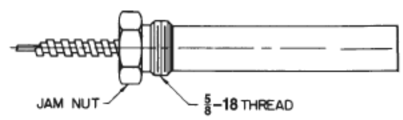
PLAIN CAPILLARY
CAT. NO. 4,6,9 SYSTEM



BRAIDED CAPILLARY
CAT. NO. 5,7 SYSTEM

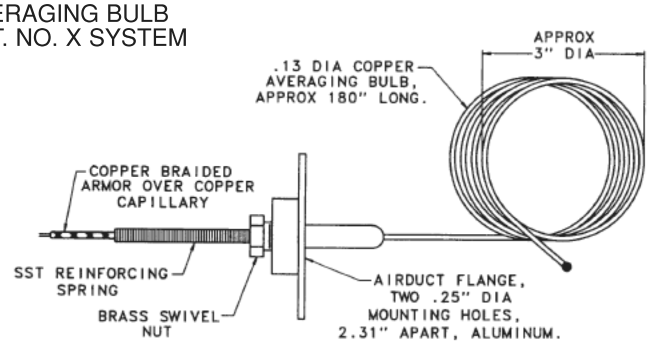


ARMORED CAPILLARY
CAT. NO. 8 SYSTEM



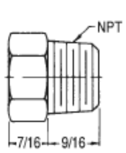
Code VI

AVERAGING BULB
CAT. NO. X SYSTEM

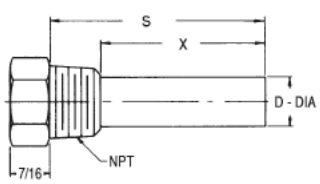


Process Connection Fittings - Code VII

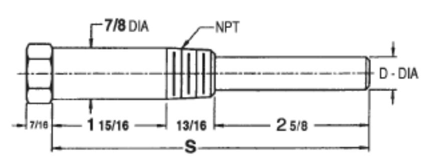
UNION CONNECTION
CAT. NO. 1,2



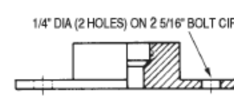
SOCKET CONNECTION
CAT. NO. 3,4



SOCKET W/ LAG EXTENSION
CAT. NO. 5,6 (No. 3 Bulb only)



FLANGE CONNECTION
CAT. NO. 7

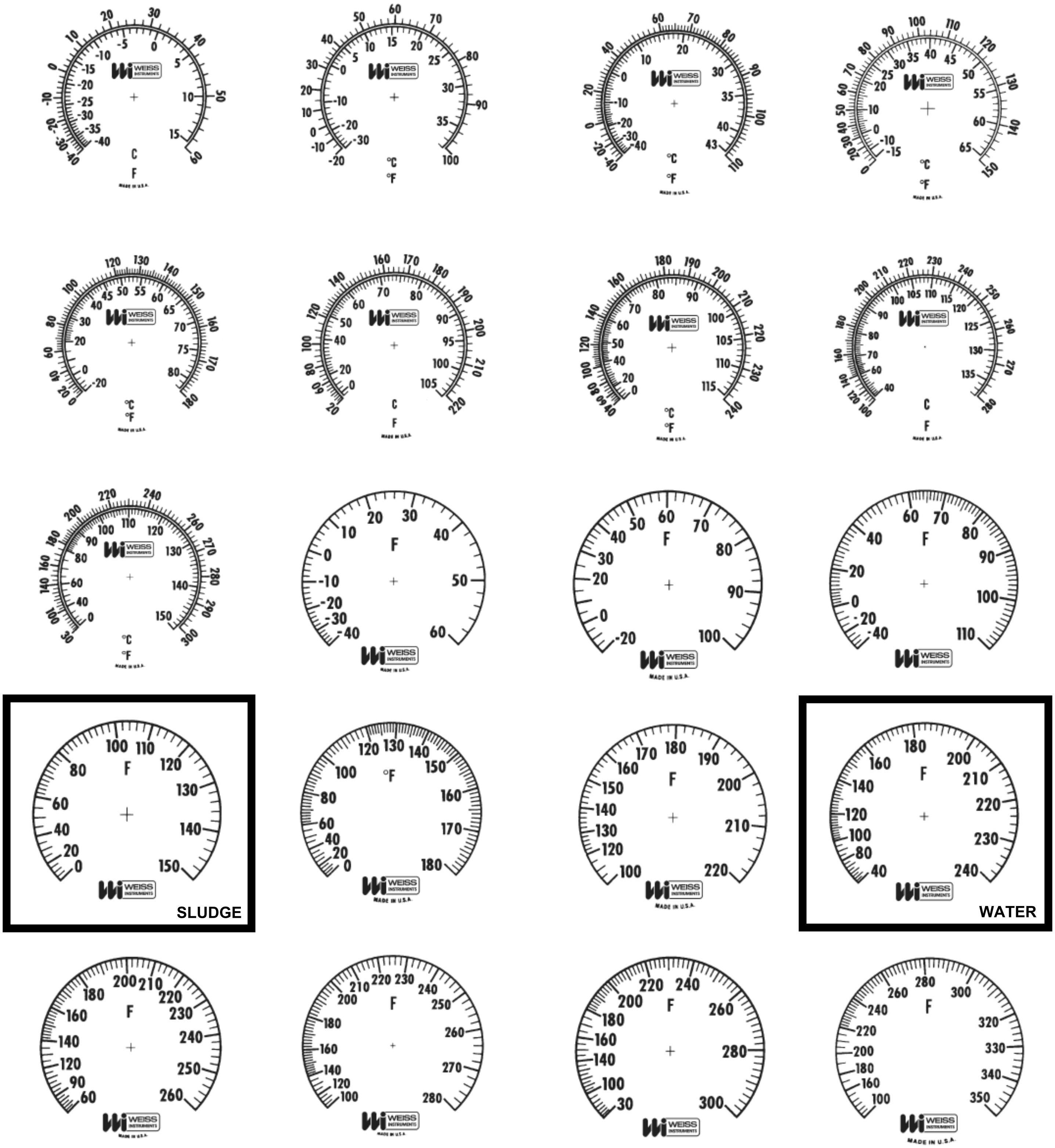


Bulb Dimensions

BULB NO. (CODE VI)	PLAIN CONNECTION		UNION CONNECTION			SOCKET CONNECTION			FLANGE CONNECTION	
	L	D	X	S	D	X	S	D	X	D
0	N/A	N/A	1	1 7/8	7/16	N/A	N/A	N/A	N/A	N/A
1	N/A	N/A	1 13/16	2 3/8	7/16	1 15/16	2 1/2	9/16	2 3/16	7/16
2	3 7/16	3/8	2 5/8	3 3/8	7/16	2 5/8	3 3/8	9/16	3	7/16
3	4 7/8	3/8	4 5/8	5 3/8	7/16	4 5/8	5 3/8	9/16	5	7/16
4	7 7/8	3/8	6 5/8	7 3/8	7/16	6 5/8	7 3/8	9/16	7	7/16
5	9 7/16	3/8	8 5/8	9 3/8	7/16	8 5/8	9 3/8	9/16	9	7/16
6	2 1/2	3/8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
7	2 1/2	5/16	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

NOTE: "X" for Flange Conn. is the Insertion Length from Tip of Bulb to Face of Flange.

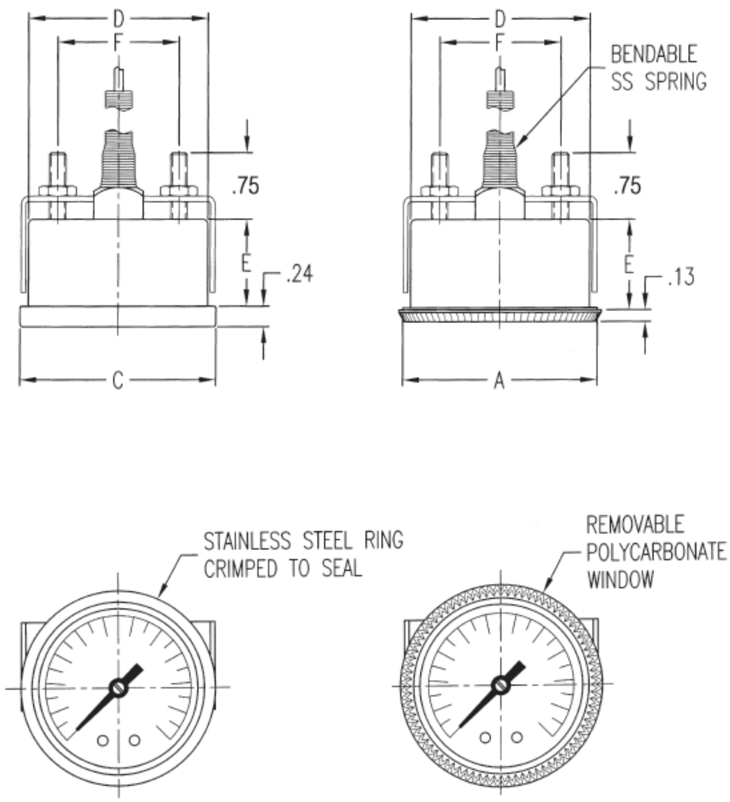
Standard Ranges
VTVA•VTR•716



NOTE: Careful consideration should be used when selecting a range, since Vapor Thermometers have progressive graduations and are best read within the upper two thirds of range. Other ranges are available, consult the factory.

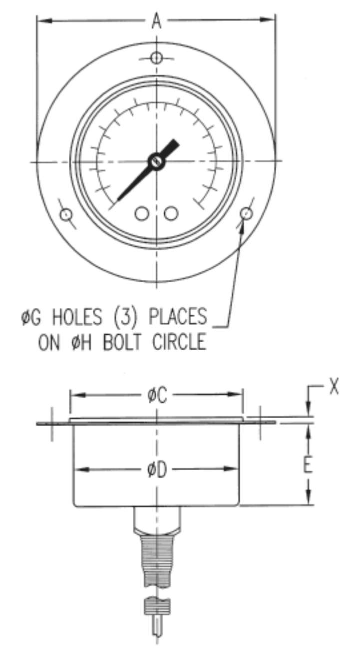
Series 716 Dimensions

Series 716 - U-Clamp Case Styles

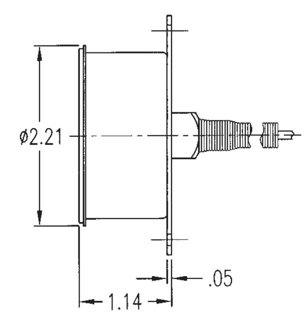


NOTE: CUSTOMER PANEL THICKNESS FOR MOUNTING FROM .03" TO .63".

Front Flange Style



Back Flange Style



Case Dimensions

CASE SIZE	CASE TYPE	CODE I	FLGE DIA. A	RING O.D. C	PANEL HOLE B	DIA. D	E	F	G	H	X CASE FRONT CODE IV		
											4	3	5
1-1/2"	U-CLAMP	15U	1-13/16	–	1-5/8	1-19/32	7/8	1-1/16	–	–	–	–	–
2"	U-CLAMP	20U	2-7/32	2-11/16	2-1/16	2-1/32	31/32	1-3/8	–	–	3/16	3/32	–
2"	FRONT FLGE	20F	2-15/16	2-11/64	2-1/16	2-1/32	31/32	–	9/64	2-9/16	3/16	3/32	–
2-1/2"	U-CLAMP	25U	2-13/16	2-7/8	2-39/64	2-9/16	1	2	–	–	3/16	1/4	–
2-1/2"	FRONT FLGE	25F	3-11/16	2-7/8	2-39/64	2-9/16	1	–	11/64	3-5/16	3/16	1/4	–
2-1/2"	FRONT FLGE	25S	3-11/16	2-7/8	2-21/32	2-41/64	3/4	–	7/32	3-5/16	3/8	–	–
3-1/2"	FRONT FLGE	35S	4-3/4	3-31/32	3-5/8	3-19/32	3/4	–	5/32	4-5/16	7/16	–	–
3-1/2"	U-CLAMP	35J	3-31/32	3-31/32	3-5/8	3-19/32	3/4	3-1/16	–	–	7/16	–	–
3-1/2"	U-CLAMP	35Q	3-7/8	3-29/32	3-5/8	3-19/32	1	3-1/16	–	–	–	–	1/8

MK 11-31,32 PRESSURE GAUGES

ASHCROFT®

Type 1279 Duragauge® Pressure Gauge Available With *PLUS!*™ Performance Option

- ***Solid front case design, field convertible to hermetically sealed or liquid filled style***
- ***Pressure ranges from vacuum — 30,000 psi.***
- ***Select from various socket and Bourdon tube materials.***
- ***Micrometer adjustable pointer.***
- ***400 Series stainless steel movement wears better for longer life***
- ***Teflon-coated pinion for longer life***
- ***Patented Duratube™ with welded-tube construction controls stresses for longer life***
- ***PLUS!™ Performance Option:***
  - ***Liquid-filled performance in a dry gauge***
  - ***Fights vibration and pulsations without liquid-filled headaches***
  - ***Order as option XLL***

The Ashcroft® Duragauge® pressure gauge is the finest production gauge on the market for industrial use where precise indications are required. The product line offers a wide variety of case styles, Bourdon tubes and pressure ranges to meet your application needs .

With the component combinations available in the Duragauge gauge line, over ten million variations are possible to serve the needs of all types of industries, including process, power, nuclear, aerospace and cryogenics.

PRODUCT SPECIFICATIONS

Model Number: 1279

Accuracy: ½% full scale (Grade 2A, ASME B40.100)

Ranges: Vac., compound to 30,000 psi

Dial Size: 4½″ diameter

Case Material: Black phenolic, solid front

Weather Protection: Dry Case: IP54
Liquid filled or hermetically sealed case: IP 65

Ring: Threaded reinforced black polypropylene

Window: Glass

Dial: Aluminum, white background, black figures and intervals.

Pointer: Micrometer adjustable

Movement: Rotary, 400 SS, Teflon® coated pinion gear and segment

Bourdon Tube and Socket: C510 Phos. bronze/brass brazed (A)
316L SS/steel (R)
316L SS/316L SS (S)
K Monel/ Monel (P)

Connection Size: ¼″, ½″ NPT

Connection Location: Lower or back

OPTIONAL FEATURES

Fill: L-Glycerin-Standard
XGV-Silicone-Optional
XGX-Halocarbon-Optional

***PLUS!* Performance:** XLL

Hermetically Sealed, IP65: H

Flush Mounting Ring: X56

Receiver Gauge: XPR

Shatter Proof Glass Window: XSG

Acrylic Window: XPD

Red Set Hand: XSH

Maximum Pointer: XEP

STANDARD RANGE TABLE*

Pressure – psi

Range	Figure interval	Minor Graduation
~~0/15~~	~~1~~	~~0.1~~
~~0/30~~	~~5~~	~~0.2~~
SLUDGE 0/60	5	0.5
WATER 0/100	10	1
~~0/160~~	~~20~~	~~2~~
~~0/200~~	~~20~~	~~2~~
~~0/300~~	~~50~~	~~2~~
~~0/400~~	~~50~~	~~5~~
~~0/600~~	~~50~~	~~5~~
~~0/800~~	~~100~~	~~10~~
~~0/1000~~	~~100~~	~~10~~
~~0/1500~~	~~200~~	~~20~~
~~0/2000~~	~~200~~	~~20~~
~~0/3000~~	~~500~~	~~20~~
~~0/5000~~	~~500~~	~~50~~
~~0/6000~~	~~500~~	~~50~~
~~0/10,000~~	~~1000~~	~~100~~
~~0/20,000~~	~~2000~~	~~200~~
~~0/30,000~~	~~5000~~	~~200~~

Compound

Range	Figure Interval		Minor Grads	
	in Hg	psi	in Hg	psi
~~30″ Hg/15 psi~~	~~5~~	~~3~~	~~0.5~~	~~0.2~~
~~30″ Hg/30 psi~~	~~10~~	~~5~~	~~1~~	~~0.5~~
SLUDGE 30″ Hg/60 psi	10	10	1	1
WATER 30″ Hg/100 psi	10	10	2	1
~~30″ Hg/150 psi~~	~~10~~	~~20~~	~~5~~	~~2~~
~~30″ Hg/200 psi~~	~~30~~	~~20~~	~~5~~	~~2~~
~~30″ Hg/300 psi~~	~~30~~	~~50~~	~~5~~	~~2~~

Vacuum

Range	Figure Interval	Minor Grads
30/0 in. Hg	5 in	0.2 in
34/0 ft H_2O	5 ft	0.5 ft

*Full standard and metric equivalent range table available on our web site.

ISO 9001
REGISTERED FIRM

BULLETIN DU-1 1279

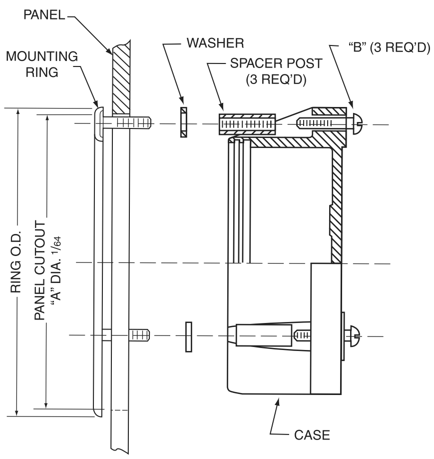
Ashcroft Inc., 250 East Main Street, Stratford, CT 06614 USA
Tel: 203-378-8281 • Fax: 203-385-0408
email: info@ashcroft.com • www.ashcroft.com

ASHCROFT®

Type 1279 Duragauge® Pressure Gauge Available With *PLUS!*™ Performance Option

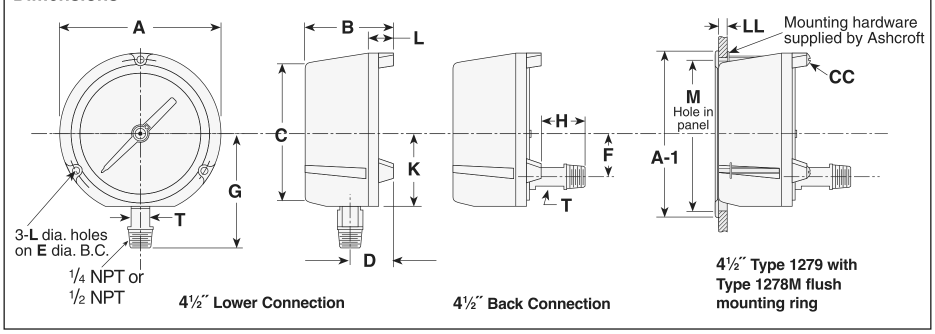
TYPE 1278M SERIES FLUSH MOUNTING RING

Used to flush mount gauge case Type 1279(*)S. Standard finish is black; polished stainless steel finish is available at an extra charge.



GAUGE SIZE	RING O.D.	"A" DIA.	"B" Size of 3 Screws	"C" Size of Washers
4½	6.000 (152)	5⅝ (148)	#10-24 x 1⅝	7/16 x 17/64 x ⅝

Dimensions



Dial Size Inches	A	B	C	D	E	F	G	H	K	L	T	V	Weight (lbs)
4½	5.81 (147.6)	3.36 (85.3)	5.07 (128.7)	1.06 (40.6)	5.375 (137)	1.62 (41.2)	4.08 (103.7)	.73 (18.4)	2.62 (66.6)	.22 (5.5)	.62 (15.7)	2.625 (67)	2½

Table A - Case selection & mounting

Dial Size (in)	Ordering Code	Case Type	Case: Material Finish	Ring: Style Material Finish	Mounting / Connection	
4½	(45)	1279(1)	Phenolic (Black)	Threaded Reinforced Polypropylene - Black	Stem –	Lower or back
					Surface –	Lower or back
					Flush –	Back: Specify X56

	Table B - System Location and Connection Bourdon Tube & Tip Material(2) (all joints TIG welded except code "A")	Socket Material(2)	Tube & Socket Code	Case Style Code	NPT Conn. & Code	Conn. Location & Code	Range Selection Limits (psi)
WATER	C510 Grade A Phosphor Bronze Tube, Brass Tip, Silver Brazed	Brass	(A)	(S) Solid Front	(02) ¼	Lower (L)	Vac./1000
	316 stainless steel	1018 steel	(R)	(S) Solid Front			Vac./20,000
SLUDGE	316 stainless steel	316 SS	(S)	(S) Solid Front	(04) ½(4)	Back (B)	Vac./20,000
	K 500 Monel®(3)	Monel 400	(P)	(S) Solid Front			Vac./30,000

NOTES:

(1) Liquid-fillable or hermetically sealed when kit 101A202-01 (lower) or kit 101A203-01 (back) is ordered.
(2) For selection of the correct Bourdon system material see the media application table.
(3) Use on applications where NACE MR-01-75 is specified for selection of the correct Bourdon system material.
(4) 30,000 psi range supplied with ¼ high pressure connection, ½ NPT optional.

HOW TO ORDER:

45 1279 RS* 04L XXX 0/2000 psi

- **From Table A, Dial Size: 4½″:** 45
- **From Table A, Case Type Number: 1279** 1279
- **From Table B, Bourdon tube and socket:** RS*
- **From Table B, Connection: ¼ NPT (02), ½ NPT (04), Lower (L), Back (B)** 04L
- **From Optional Features:** XXX
- **From Range Table: 2000 psi** 0/2000 psi

(*) "S" denotes solid front case design
If glycerin filled case, substitute "L" instead of "S"

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MK 32-1 & 33-1

PRESSURE RELIEF VALVES

F-82 Safety Relief Valve

DESCRIPTION

The Cash Acme F-82 is a pressure-only ASME safety relief valve designed for use on hot water space heating boilers, water supply heaters and storage tanks.

The Cash Acme F-82 Safety Relief Valves are designed for use on hot water service only and are not to be used on steam. The Cash Acme F-82 is available in 3/4", 1", 1-1/4", 1-1/2" and 2" sizes. All sizes have a female inlet and outlet. The 3/4" size is also available with a male inlet.

The Cash Acme F-82 ASME Safety Relief Valves have a standard factory pressure relief setting from 30 to 150 psi. These high capacity pressure-only safety relief valves feature bronze bodies, silicone seat discs, EP diaphragms, brass internal parts, and stainless steel pressure springs.

Lead Free* versions of most models are also available.

FEATURES AND BENEFITS

Compact and economical:
Saves space and money!

Completely automatic and reseats after pressure relief:
Ideal for most domestic water heater applications.

ASME Section IV listed:
Rated for emergency steam discharge by the National Board of Boiler and Pressure Vessel Inspectors.

Every valve is tested for performance prior to shipping:
Specify and install with confidence!

SPECIFICATION

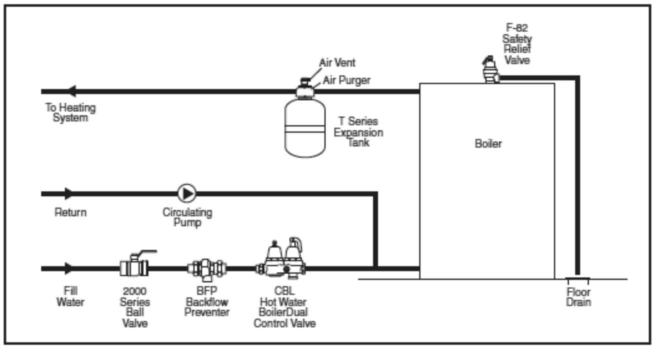
A high capacity safety relief valve shall be installed to relieve pressure at a pre-determined set. The valve shall have a bronze body with a silicone seat disc and stainless steel pressure spring. The valve shall be ASME listed and National Board Certified. The valve shall be a Cash Acme F-82.

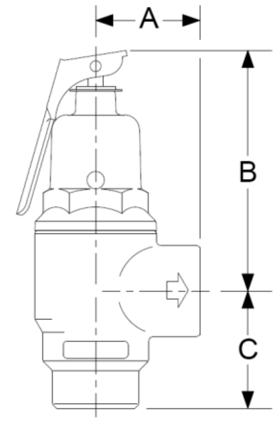
* A device is defined as Lead Free if its normally wetted surface has a weighted average lead content not exceeding 0.25%.

CASH ACME

F-82 Safety Relief Valve

TYPICAL INSTALLATION





Inlet Size (NPS) Female	Outlet Size (NPS) Female	DIMENSIONS		
		A	B	C
3/4"	3/4"	1-1/2"	3-1/2"	1-11/16"
1"	1"	1-3/4"	4-11/32"	1-7/64"
1-1/4"	1-1/4"	2-1/16"	5-9/16"	1-1/2"
1-1/2"	1-1/2"	2-15/32"	5-7/8"	1-21/32"
2"	2"	3"	7-1/2"	2-5/32"

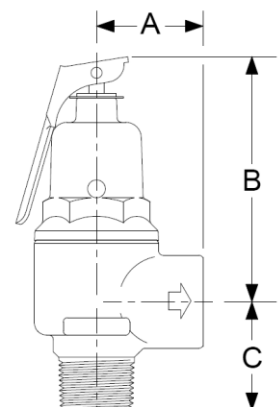
SPECIFICATION DATA

Performance:
Available Set pressures ……………………… 30-150 psi
Service ………………………………………… Water

Materials:
Body …………………………………………... Bronze
Internal Parts ………………………………… Brass
Seat Disc ……………………………………… Silicone
Pressure Spring ……………………………… Stainless steel
Diaphragm …………………………………… EP

AVAILABLE CONNECTIONS

Threaded (NPT) ……………………………… Female inlet and female outlet
(Male inlet also available on 3/4" size)



Inlet Size (NPS) Male	Outlet Size (NPS) Female	DIMENSIONS		
		A	B	C
3/4"	3/4"	1-1/2"	3-1/2"	1-17/32"

CERTIFICATIONS

The Cash Acme F-82 Safety Relief Valve is capacity certified to ASME Boiler Code Section IV.

ASME RELIEF CAPACITIES AT STANDARD SETTINGS - BTU/HR STEAM RELIEF					
RELIEF SETTING	F-82 3/4" x 3/4"	F-82 1" x 1"	F-82 1-1/4" x 1-1/4"	F-82 1-1/2" x 1-1/2"	F-82 2" x 2"
30psi	717,000	1,025,000	1,704,000	2,085,000	3,902,000
50psi	1,048,000	1,498,000	2,490,000	3,047,000	5,701,000
75psi	1,462,000	2,090,000	3,472,000	4,249,000	7,951,000
100psi	1,875,000	2,680,000	4,454,000	5,451,000	10,200,000
125psi	2,289,000	3,271,000	5,437,000	6,653,000	12,450,000
150psi	2,703,000	3,862,000	6,419,000	7,855,000	14,699,000